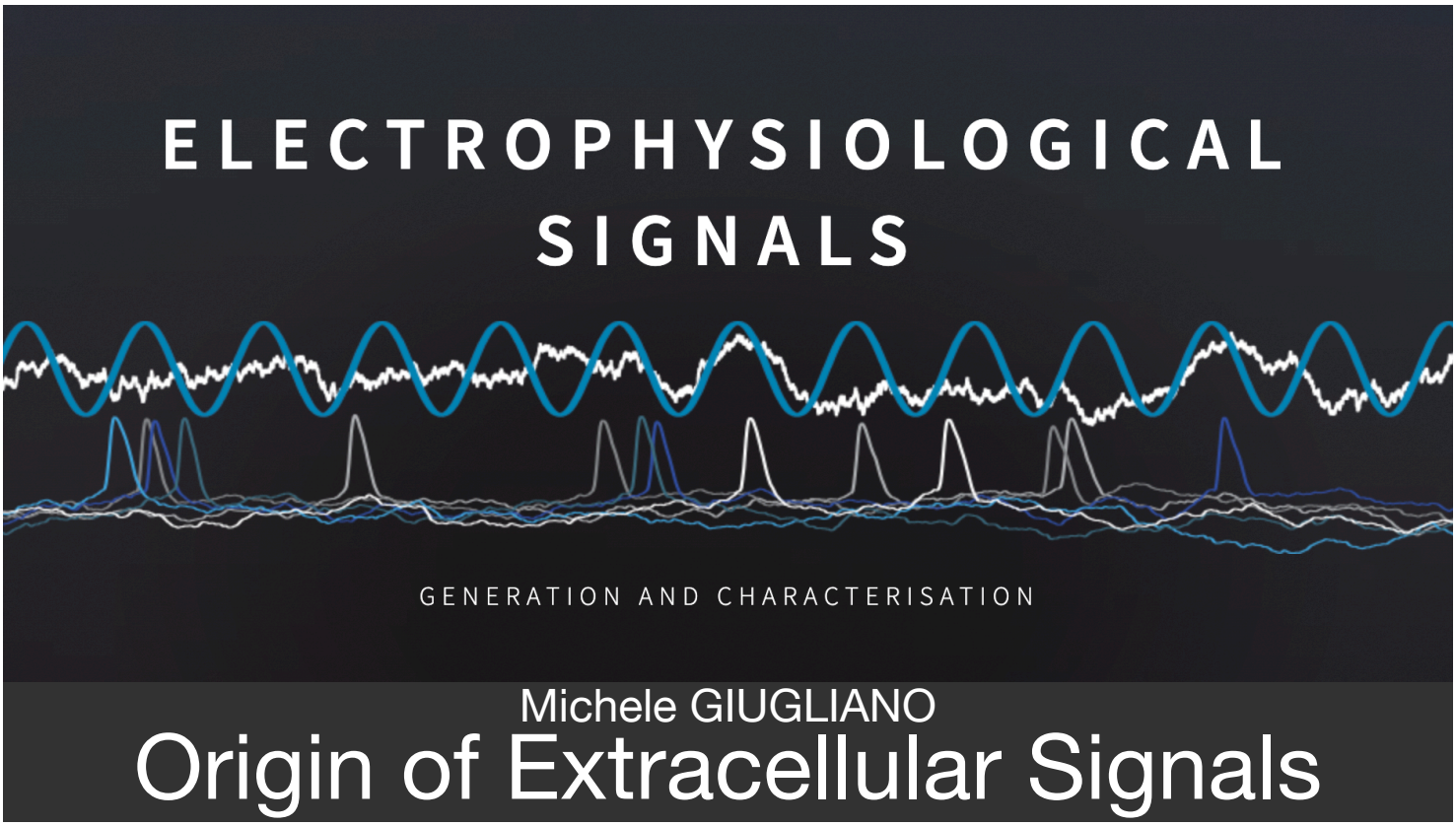


Michele GIUGLIANO

Origin of Extracellular Signals

ATTENDANCE TRACKING - **code LCON5**
(for statistical purposes only)

https://www.unimore.it/it/servizi/unimore-app

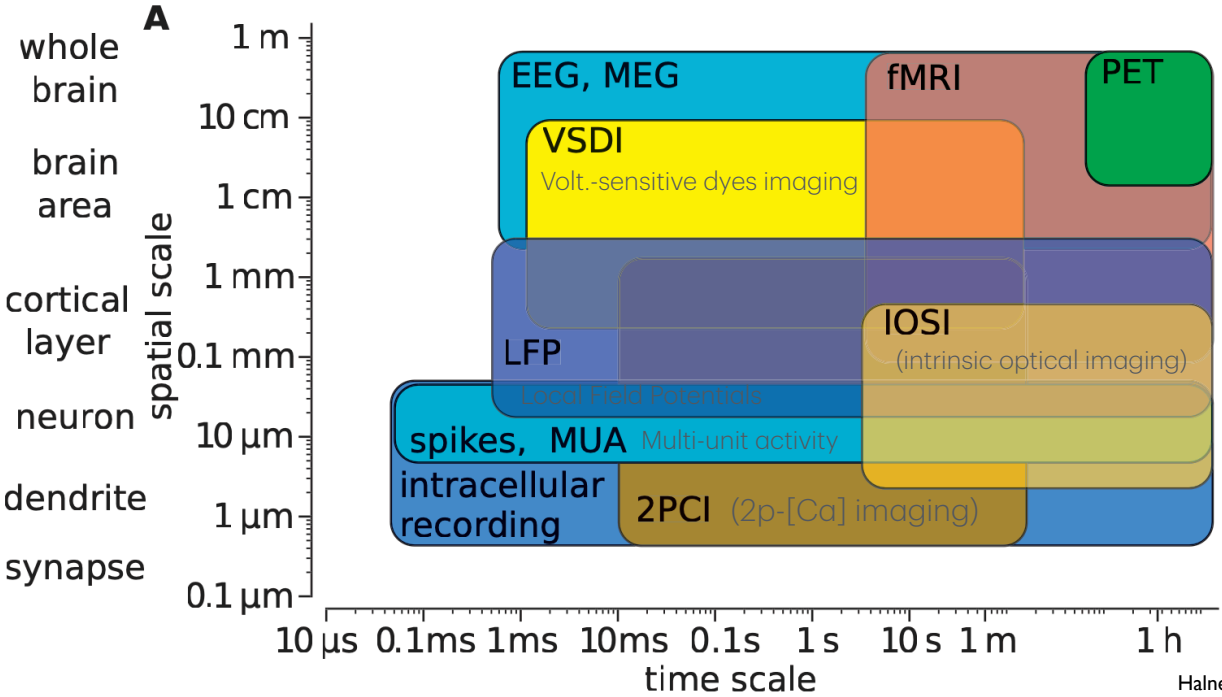
References

supporting your study and understanding

Chapters from

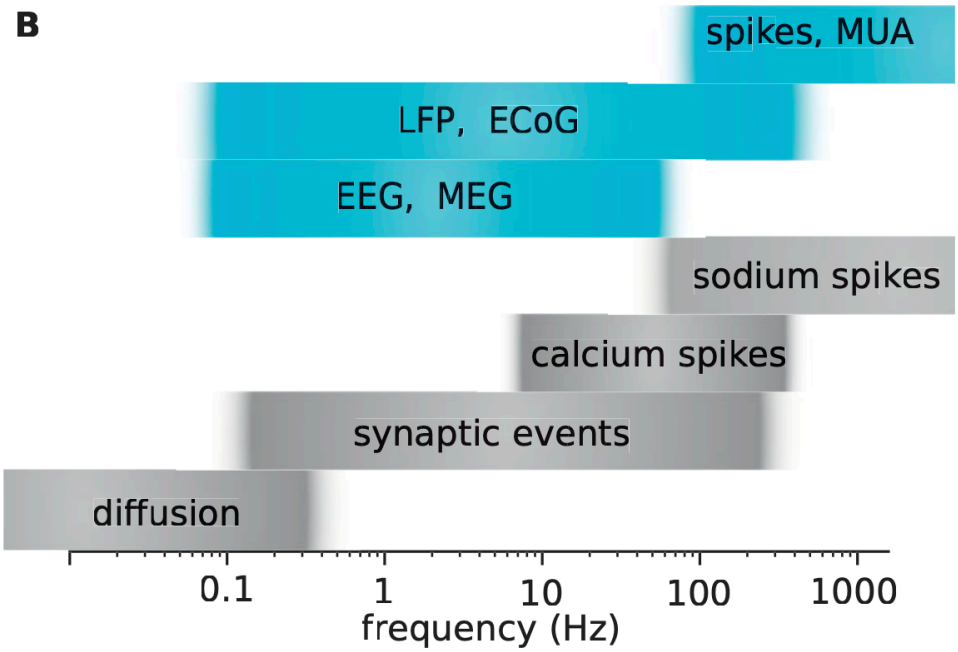
- Halnes et al., (2024) "*Electric Brain Signals*"
- Johnston & Wu (1994) *"Foundations of Cellular Neurophysiology"*
- Sterratt et al. (2011) "*Principles of Computational Modelling…*"
- Abbott LF, Dayan P (2001) "*Theoretical Neuroscience*"

Range of Techniques and Methods: not all the same!



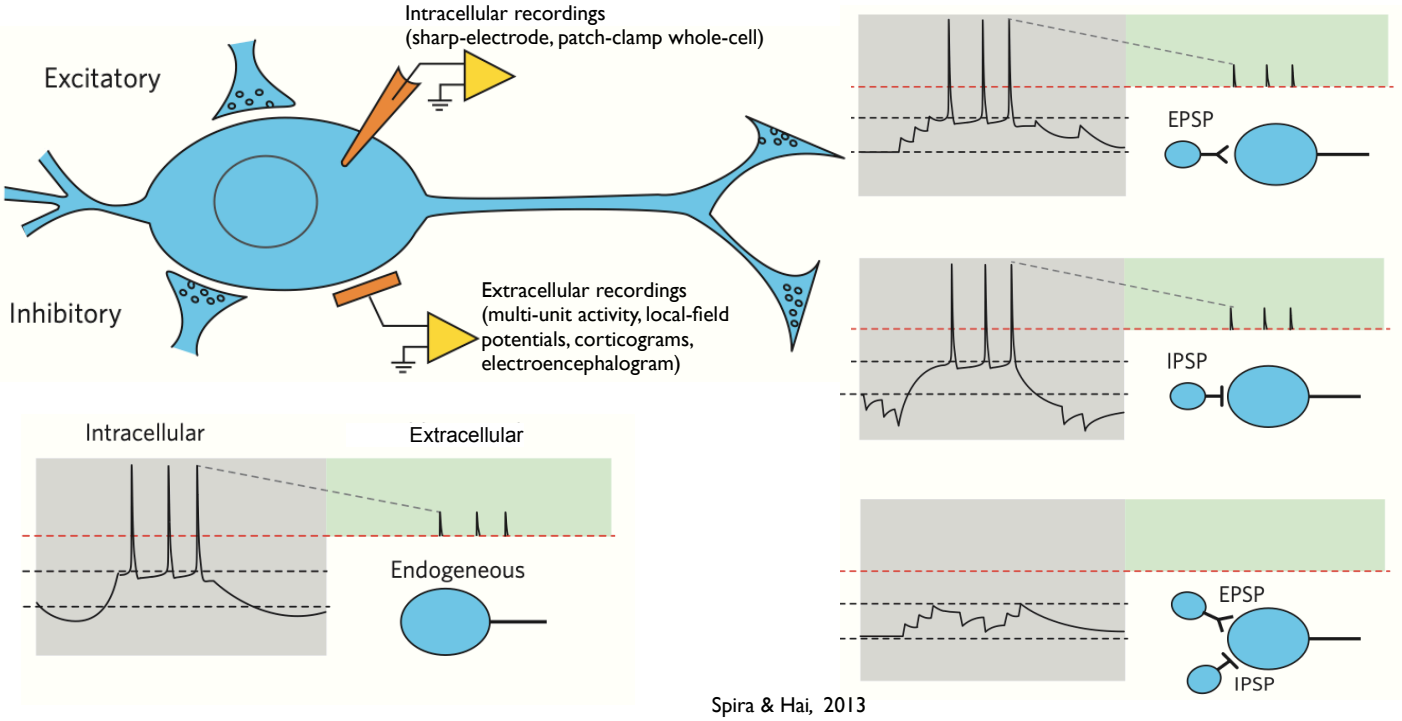
Halnes *et al.*, 2024

Heterogeneous (and multiple) time-scales



Halnes *et al.*, 2024

*Extra*cellular signals: only the *"fast/strong part"*?



Spira & Hai, 2013

*Extra*cellular signals: not so elementary!

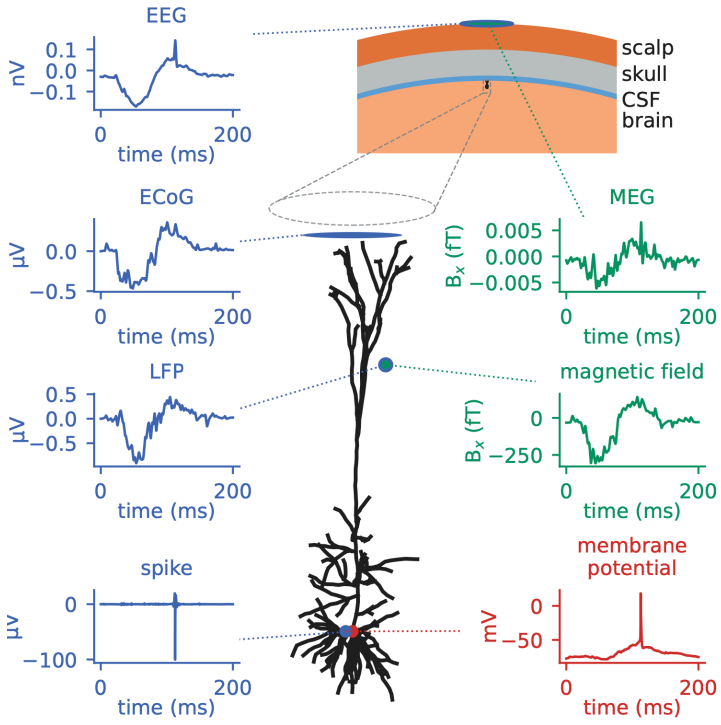
1) **Where** are you recording?

2) From **how far** in space is your recording technique sensitive too? Are you "listening" to one cell? To multiple cells?

3) How cells are **shaped**, in 3d?

4) **Where are**, if any, the synaptic **inputs** impinging to?

5) **What** (physical signals) are you recording?



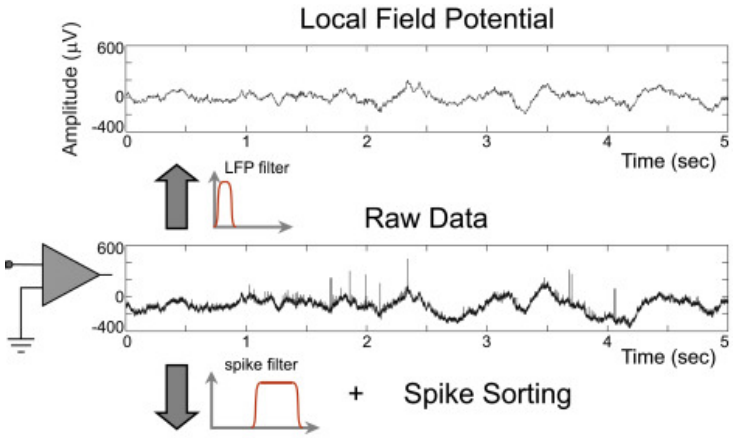
Halnes *et al.*, 2024

Low **F**requency **P**art**s**/components

1 - 100 cycle/s

LFP

Local
Field
Potential



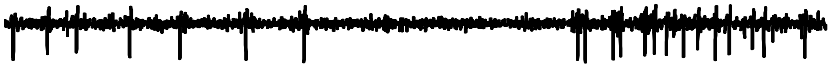
Raw recording

High-freq. components

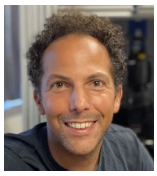
100 - 5000 cycle/s

MUA
SUA

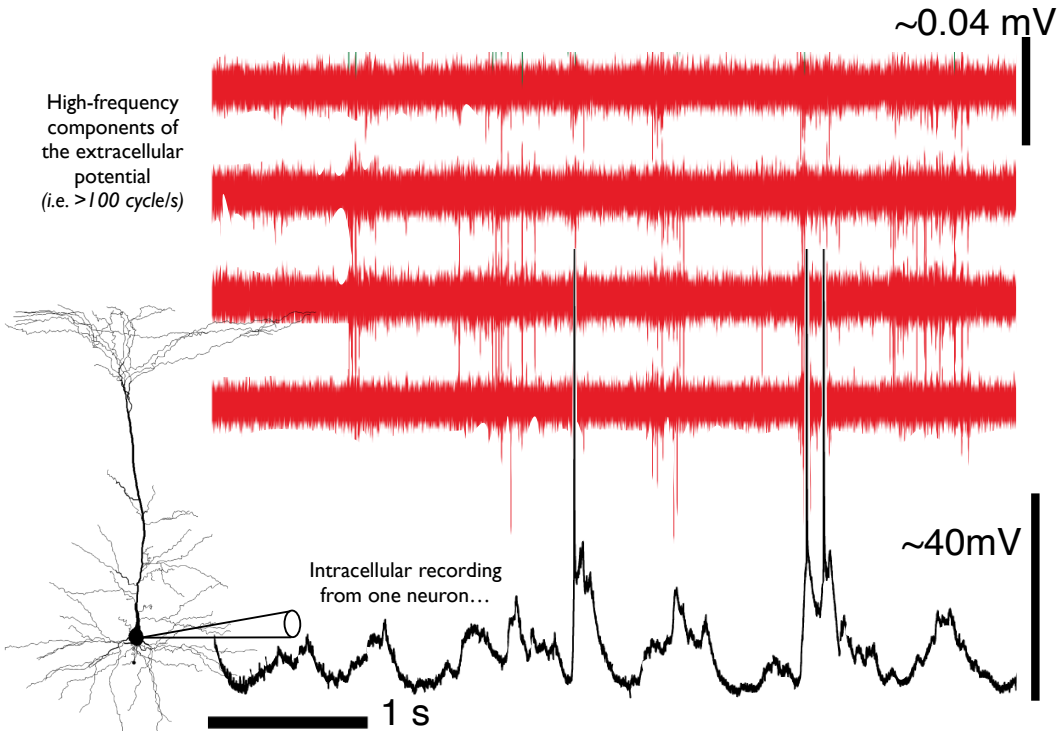
Multi-unit
Single-unit
Activity

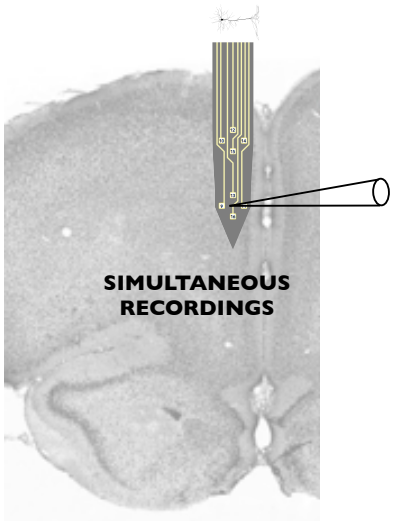


*Extra*cellular signals: APs, near the soma?



João COUTO



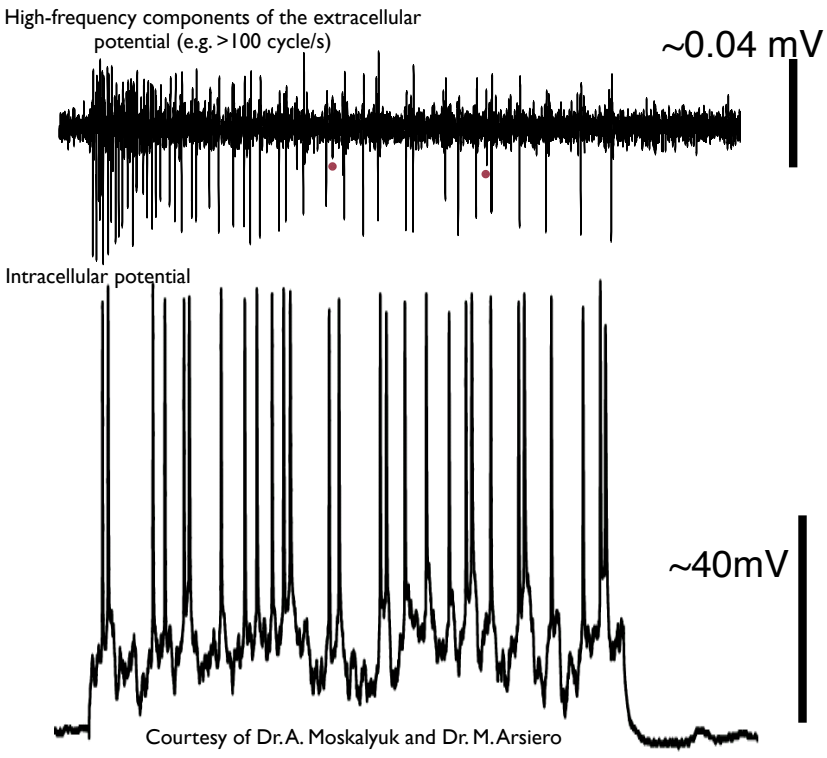


Courtesy of Dr. J. Couto

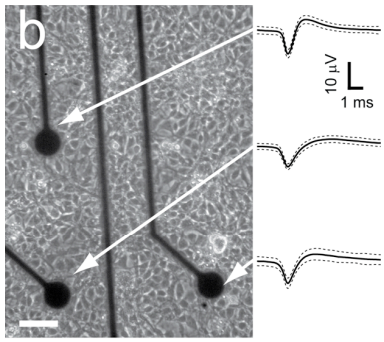
Extracellular signals: APs, near the soma?

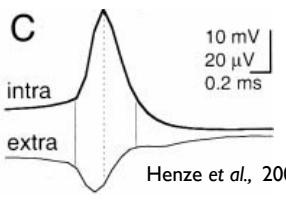
Dr. A. Moskalyuk

Dr. Eng. M. Arsiero



Courtesy of Dr. A. Moskalyuk and Dr. M. Arsiero

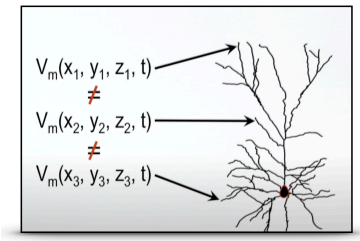


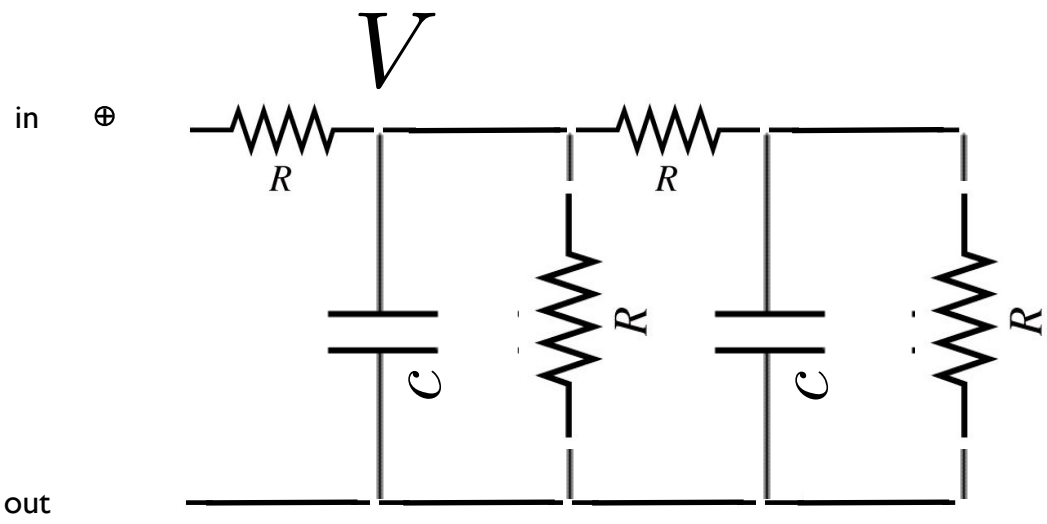


Henze *et al.*, 2000

How to make sense of this?

*Dendrites are **non**-isopotential portions of the neuron!*

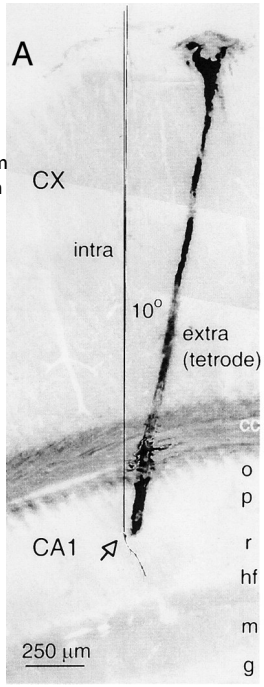


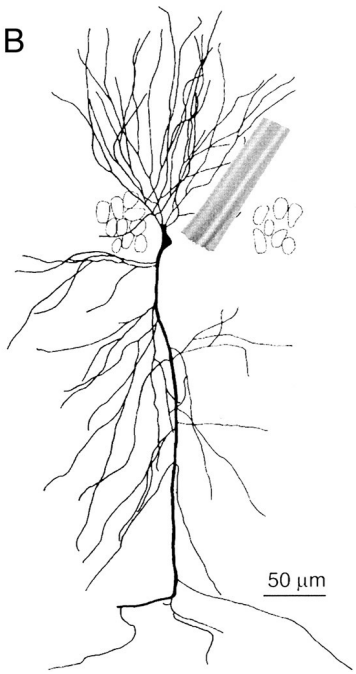


Extracellular signals: APs, near the soma

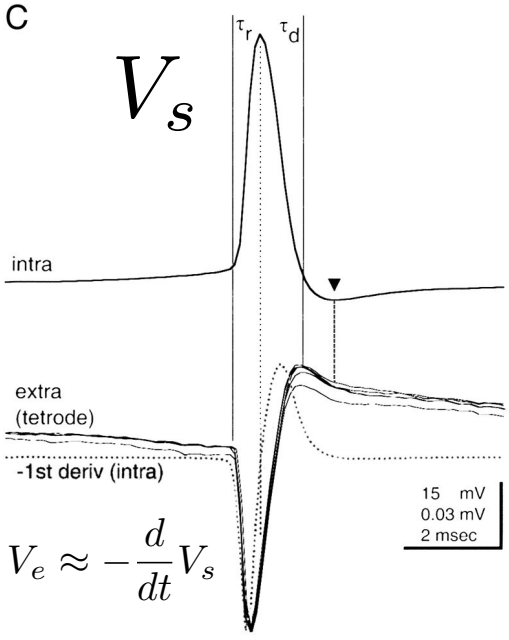
In vivo recordings from rat hippocampus, with both tungsten wire(s) and sharp electrodes.

György Buzsáki

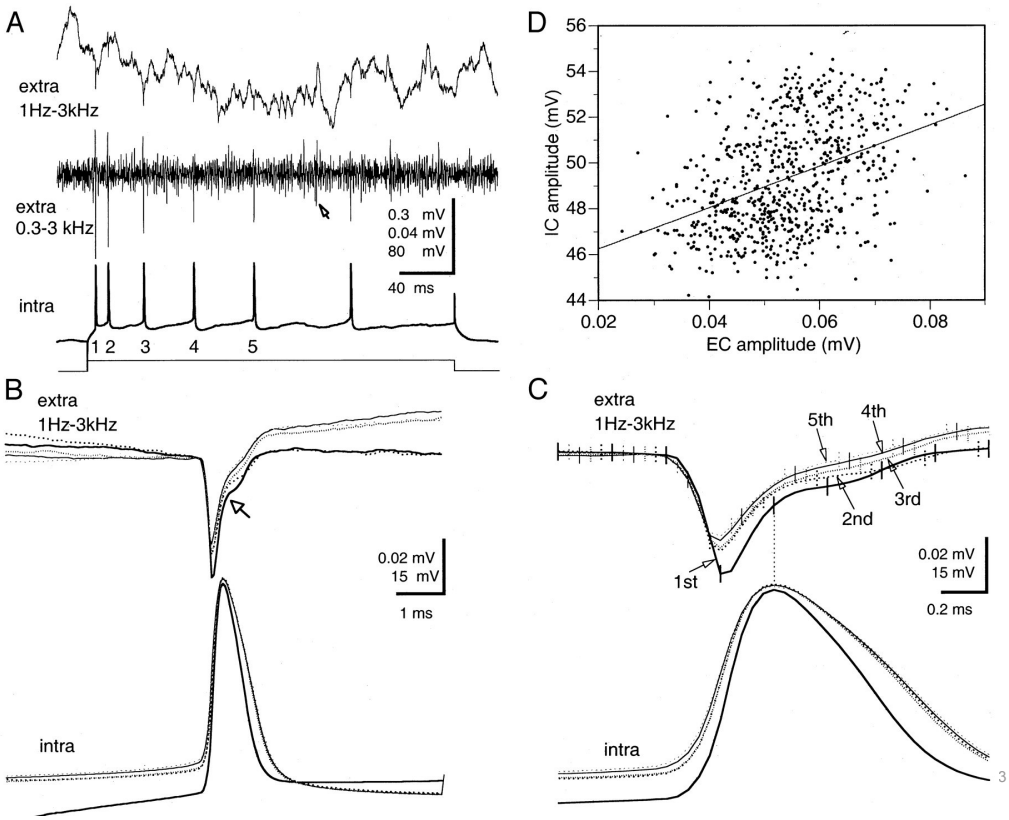




CA1, hippocampal pyramidal cell



Henze *et al.,* 2000

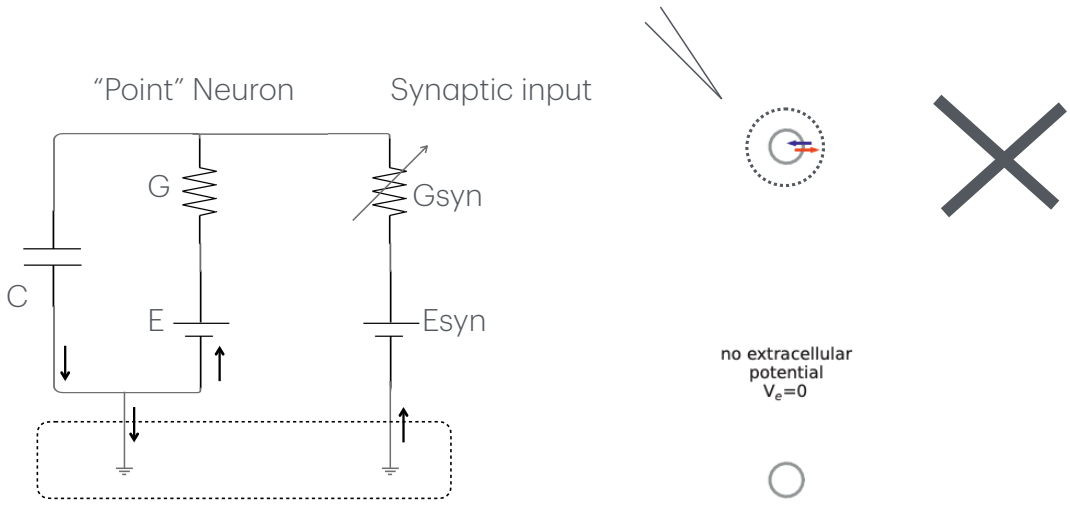


$$V_e \approx -\frac{d}{dt} V_s$$

V_e ~proportional to both **peak** AP amplitude **and slope**... V_s

Henze *et al.*, 2000

Extracellular signals: APs, near the soma?



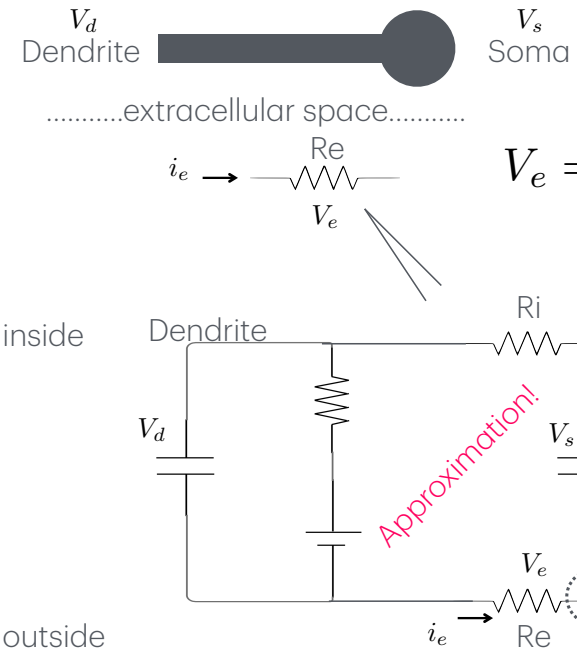
Currents, flowing in and out, **cancel** (at the same point in space).

No extracellular (difference of) potential(s) is generated.

Extracellular signals: APs, near the soma?

We must take the spatial extension of neurons into account!

$V_d \neq V_s$



$$V_e = R_e \; i_e$$

$$i_e + i_c + i_m = 0$$

$$i_e + C_s \frac{d}{dt} V_s + G_s(V_s - E_s) = 0$$

$$V_e = R_e \left(-C_s \frac{d}{dt} V_s + G_s(E_s - V_s) \right)$$

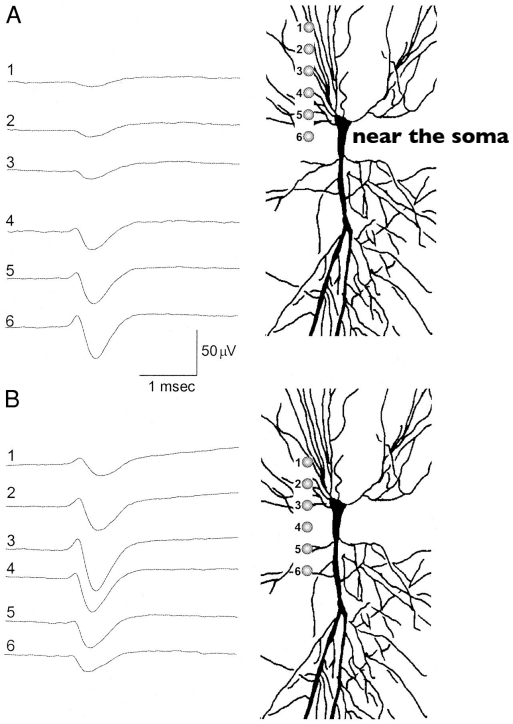
$$C_s \frac{d}{dt} V_s >> G_s(E_s - V_s)$$

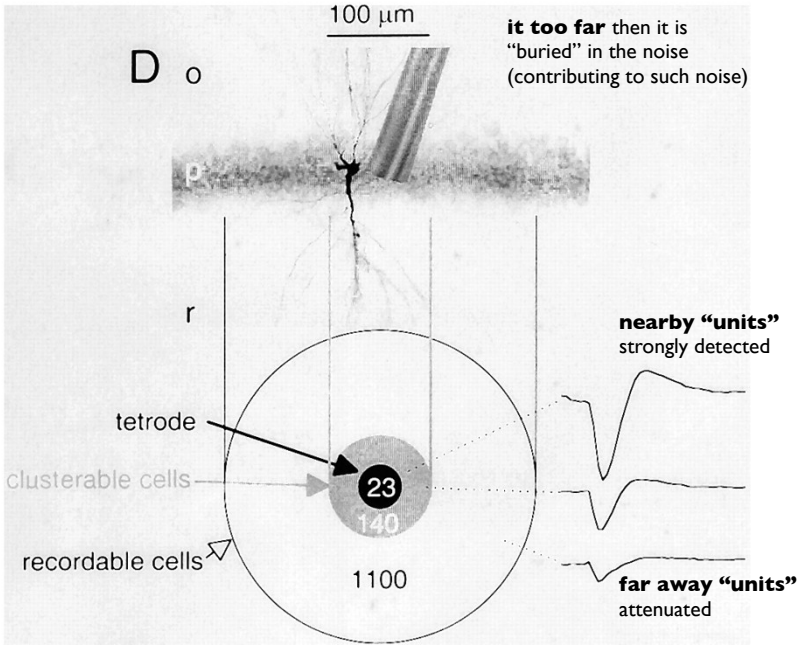
$$V_e \approx -\frac{d}{dt} V_s$$

Extracellular signals: position and distance

where are you recording?

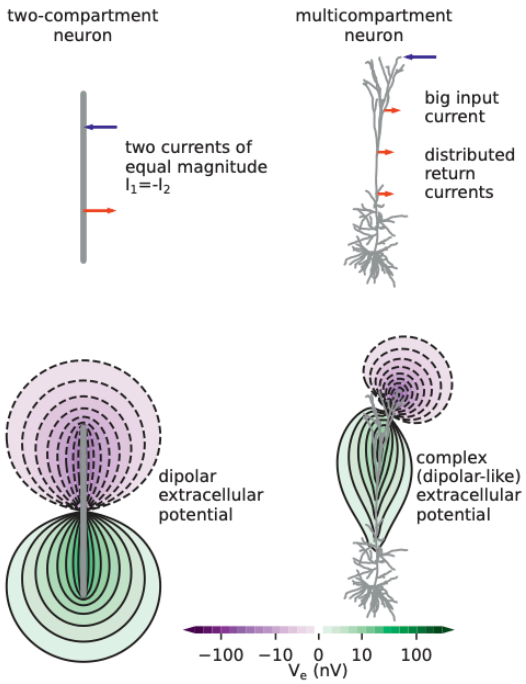
how far are you recording?





Henze *et al.*, 2000

Extracellular signals: more in general, how to?



Halnes *et al.*, 2024

- Describing, **across space**, ionic currents changes.

(**cable theory**)

$$\lambda^2 \frac{\partial^2 v(x,t)}{\partial x^2} = \tau_m \frac{\partial v(x,t)}{\partial t} + v(x,t)$$

- Computing extracellular potentials, from the **spatial distribution of transmembrane currents**.

(**volume-conductor theory**)

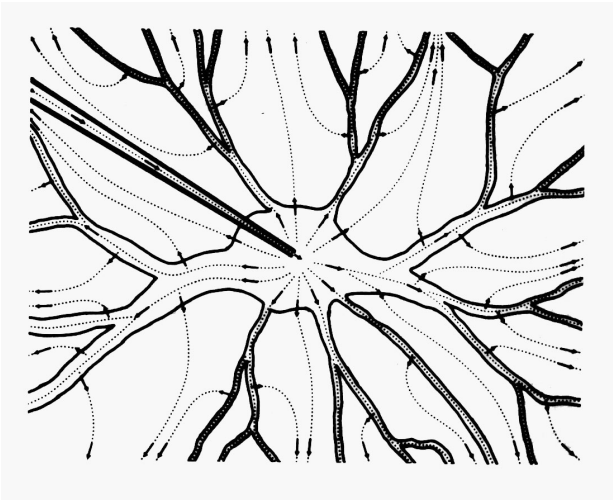
$$V_e(\mathbf{r}) = \sum_{\mathbf{n}} \frac{\mathbf{i_n}}{4\pi\sigma_\mathbf{t}|\mathbf{r} - \mathbf{r_n}|}$$

Let's put on hold our discussion on extracellular signals (and volume-conductor theory).

Let's focus on space-dependent electrical properties and electrical phenomena of neurons. (cable theory)

1959 Rall's Cable Theory for Neurites: motivations

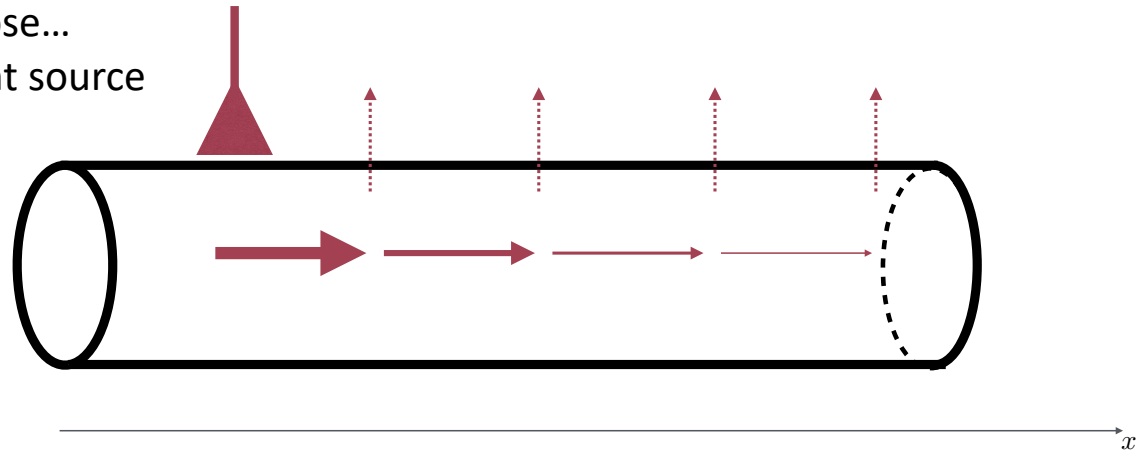
- Most of the input current flows into the dendrites (not directly into soma)!
- Anyway, current have <u>to flow outside</u> the cell for"closing the circuit".



Wilfried Rall

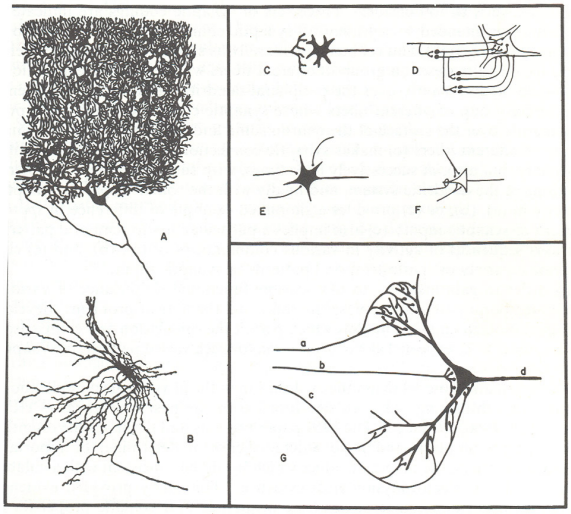
1959 Rall's Cable Theory for Neurites: intuition

synapse…
~~ current source



1959 Rall's Cable Theory for Neurites: motivations

- Understanding the **impact** of (**remote**) **dendritic** synapses activation (i.e. inputs) on the membrane potential of the **soma/axon** (i.e. output) region.



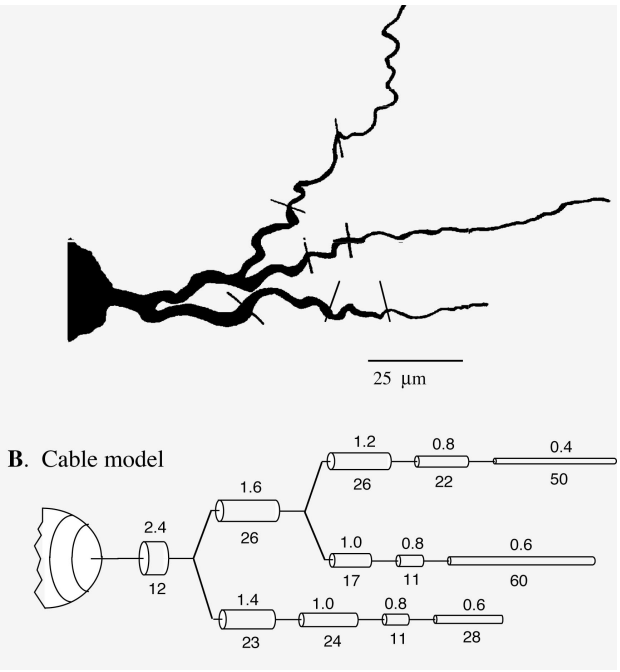
RESULTS:

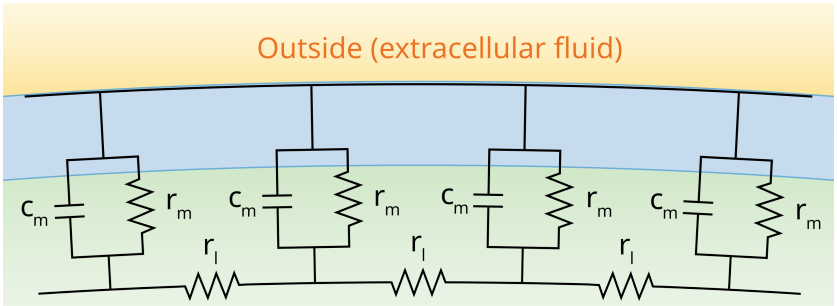
- Dendrites are non-isopotential portions of a neuron!
  - (i) voltages **attenuate** from synapse to soma!
  - (ii) it takes **time** for distal PSPs to reach the soma
  - (iii) the **shape** of EPSP/IPSP detected at the soma, **changes with** synaptic **distance**.

ATTENDANCE TRACKING - **code O1BNO**
(for statistical purposes only)

https://www.unimore.it/it/servizi/unimore-app

cylindrical symmetry, 1-d situation, thus 1-d cable!





Specific electrical properties:

Capacitance

Resistance

r_m: Membrane resistance
per unit of (lateral) area

r_l: Longitudinal resistance
per unit of (section) area and unit of length

c_m: Capacitance due to electrostatic forces
per unit of (lateral) area

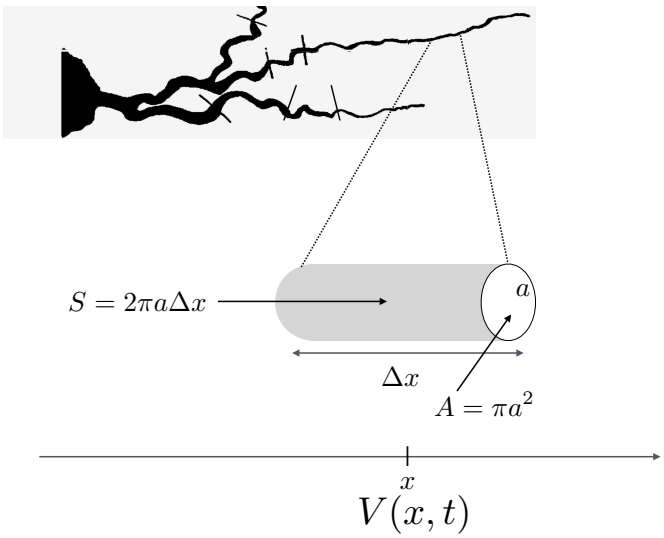
$$c = 1\mu F cm^{-2} \qquad r_m = 20k\Omega cm^2 \qquad r_i = 200\Omega cm$$

$$cr_m = 1\mu F cm^{-2} 20k\Omega cm^2 = 20ms$$

Cable Theory for Dendrites (and Axons)

Let's take an infinitesimal section…

$$V(x,t)$$

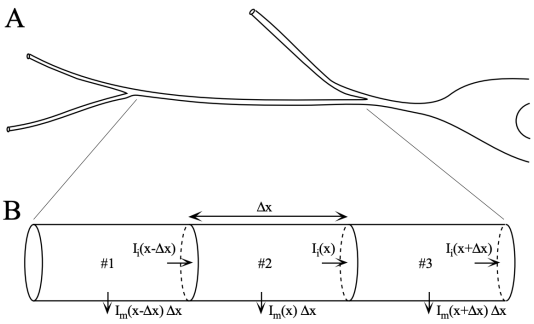


Total electrical properties:

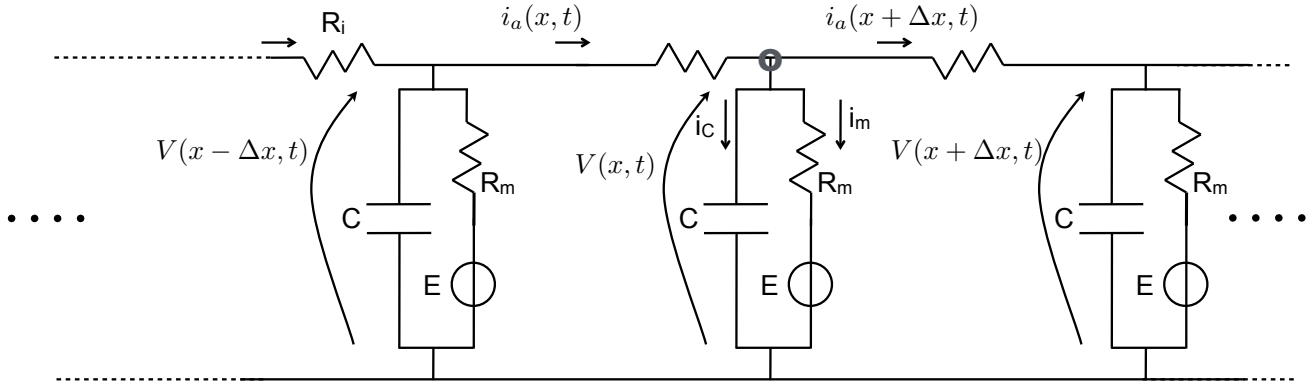
$$R_m = r_m/S = r_m/(2\pi a \Delta x) \qquad r_m = 20k\Omega cm^2$$

$$C = cS = c(2\pi a \Delta x) \qquad c = 1\mu F cm^{-2}$$

$$R_i = r_i \Delta x/(a^2\pi) \qquad r_i = 200\Omega cm$$



Cable Theory for Dendrites (and Axons)



$$i_a(x,t) = i_C(x,t) + i_m(x,t) + i_a(x+\Delta x,t)$$

$$i_a(x,t) = \frac{V(x-\Delta x,t) - V(x,t)}{R_i} \qquad i_m(x,t) = \frac{V(x,t) - E}{R_m}$$

$$i_a(x+\Delta x,t) = \frac{V(x,t) - V(x+\Delta x,t)}{R_i} \qquad i_C(x,t) = C\frac{dV(x,t)}{dt}$$

$$\frac{V(x-\Delta x,t) - V(x,t)}{R_i} = C\frac{dV(x,t)}{dt} + \frac{V(x,t) - E}{R_m} + \frac{V(x,t) - V(x+\Delta x,t)}{R_i}$$

Cable Theory for Dendrites (and Axons)

$$\frac{V(x-\Delta x,t) - V(x,t)}{R_i} = C\frac{dV(x,t)}{dt} + \frac{V(x,t) - E}{R_m} + \frac{V(x,t) - V(x+\Delta x,t)}{R_i}$$

$$v(x,t) = V(x,t) - E$$

$$\frac{v(x-\Delta x,t) - 2v(x,t) + v(x+\Delta x,t)}{R_i} = C\frac{dv(x,t)}{dt} + \frac{v(x,t)}{R_m}$$

$$\frac{R_m}{R_i}\left(v(x-\Delta x,t) - 2v(x,t) + v(x+\Delta x,t)\right) = R_m C\frac{dv(x,t)}{dt} + v(x,t)$$

$$v(x-\Delta x,t) \approx v(x,t) - \Delta x \ \frac{dv(x,t)}{dx} + \frac{1}{2}\Delta x^2 \ \frac{d^2v(x,t)}{dx^2} \quad +$$

$$v(x+\Delta x,t) \approx v(x,t) + \Delta x \ \frac{dv(x,t)}{dx} + \frac{1}{2}\Delta x^2 \ \frac{d^2v(x,t)}{dx^2}$$

$$v(x-\Delta x,t) + v(x+\Delta x,t) \approx 2v(x,t) + \Delta x^2 \ \frac{d^2v(x,t)}{dx^2}$$

…Invoking Taylor's…

$$\frac{R_m}{R_i}\left(v(x-\Delta x,t) - 2v(x,t) + v(x+\Delta x,t)\right) = R_m C \frac{dv(x,t)}{dt} + v(x,t)$$

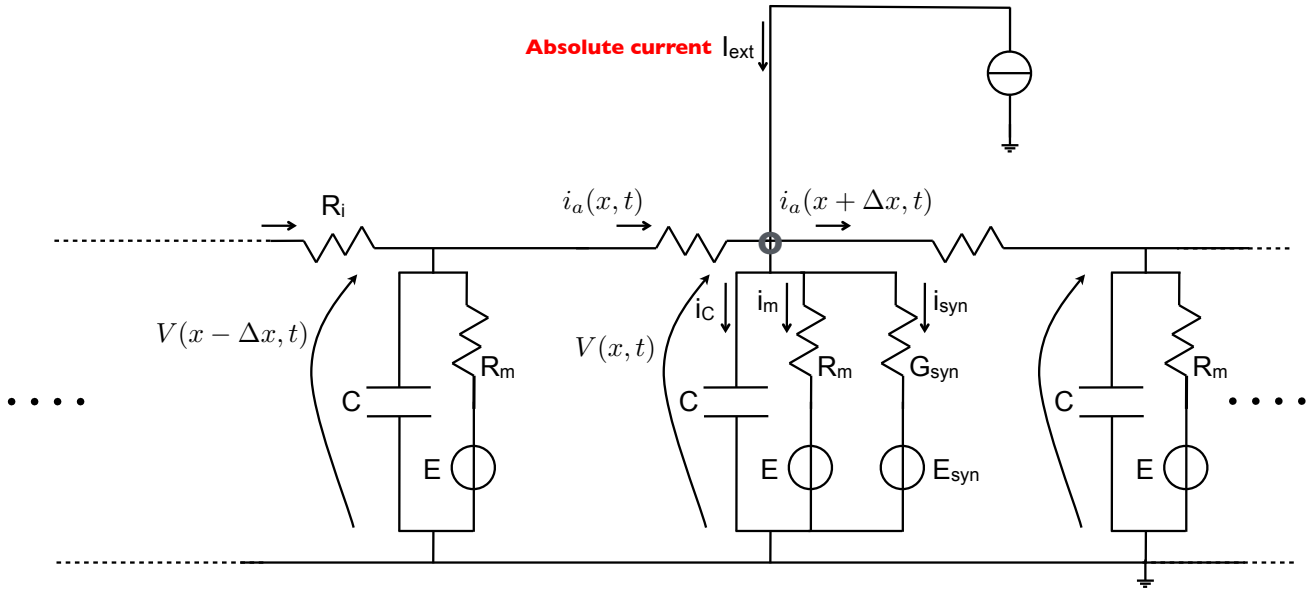
$$\frac{R_m}{R_i}\left(\Delta x^2 \frac{d^2v(x,t)}{dx^2}\right) = R_m C \frac{dv(x,t)}{dt} + v(x,t)$$

PDE!

$R_m = r_m/S = r_m/(2\pi a \Delta x)$
$C = cS = c(2\pi a \Delta x)$
$R_i = r_i \Delta x/(a^2\pi)$

$$\frac{a}{2}\frac{r_m}{r_i}\frac{\partial^2 v(x,t)}{\partial x^2} = r_m c \frac{\partial v(x,t)}{\partial t} + v(x,t)$$

…with synaptic and external current *densities*.



$$i_a(x,t) + I_{ext}(x,t) = i_C(x,t) + i_m(x,t) + i_{syn}(x,t) + i_a(x+\Delta x,t)$$

$$i_{syn}(x,t) = G_{syn}\ (V(x,t) - E_{syn})$$

$$i_{ext} = \frac{I_{ext}(x,t)}{2\pi a \Delta x}$$

$$\lambda^2 \frac{\partial^2 v(x,t)}{\partial x^2} = \tau_m \frac{\partial v(x,t)}{\partial t} + v(x,t)$$

$$\tau_m = r_m c$$

$$\lambda = \sqrt{\frac{a}{2}\frac{r_m}{r_i}}$$

$$\lambda^2 \frac{\partial^2 V(x,t)}{\partial x^2} = \tau_m \frac{\partial V(x,t)}{\partial t} + (V(x,t) - E)$$

$$\lambda^2 \frac{\partial^2 V(x,t)}{\partial x^2} = \tau_m \frac{\partial V(x,t)}{\partial t} + (V(x,t) - E) + r_m \ [g_{syn}(x,t)(V(x,t) - E_{syn}) - i_{ext}(x,t))]$$

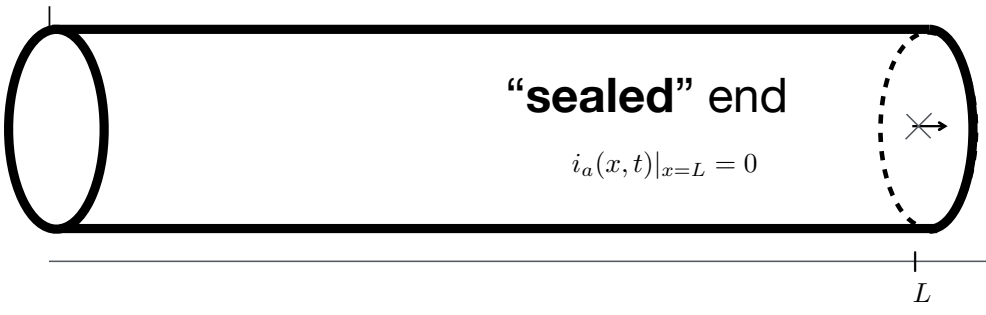
2 x Boundary conditions
1 x Initial condition
needed

$$i_{ext} = \frac{I_{ext}(x,t)}{2\pi a \Delta x}$$

Boundary conditions: 1

$$i_a(x,t) = \frac{v(x-\Delta x,t) - v(x,t)}{R_i} \qquad i_a(x+\Delta x,t) = \frac{v(x,t) - v(x+\Delta x,t)}{R_i}$$

$$i_a(x,t)|_{x=L} = \frac{a^2\pi}{r_i} \frac{v(x,t) - v(x+\Delta x,t)}{\Delta x}|_{x=L} = -\frac{a^2\pi}{r_i} \frac{d}{dx} v(x,t)|_{x=L}$$

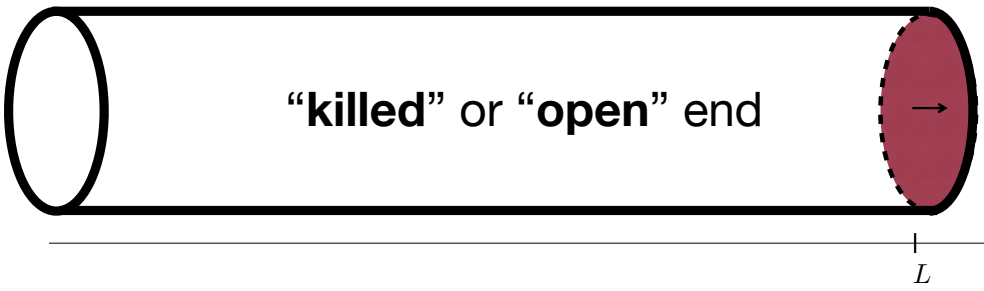


$$\frac{d}{dx} v(x,t)|_{x=L} = 0$$

"open-circuit" in electrical *jargon*!!

Boundary conditions: 2

$$V_{in}(L,t) = V_{out} = 0$$



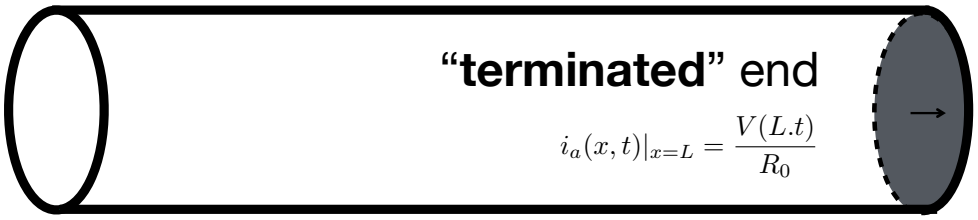
$$V(L,t) = 0$$

L

"short-circuit" in electrical *jargon*!!

Boundary conditions: 3

$$i_a(x,t) = \frac{v(x-\Delta x,t) - v(x,t)}{R_i} \qquad i_a(x+\Delta x,t) = \frac{v(x,t) - v(x+\Delta x,t)}{R_i}$$

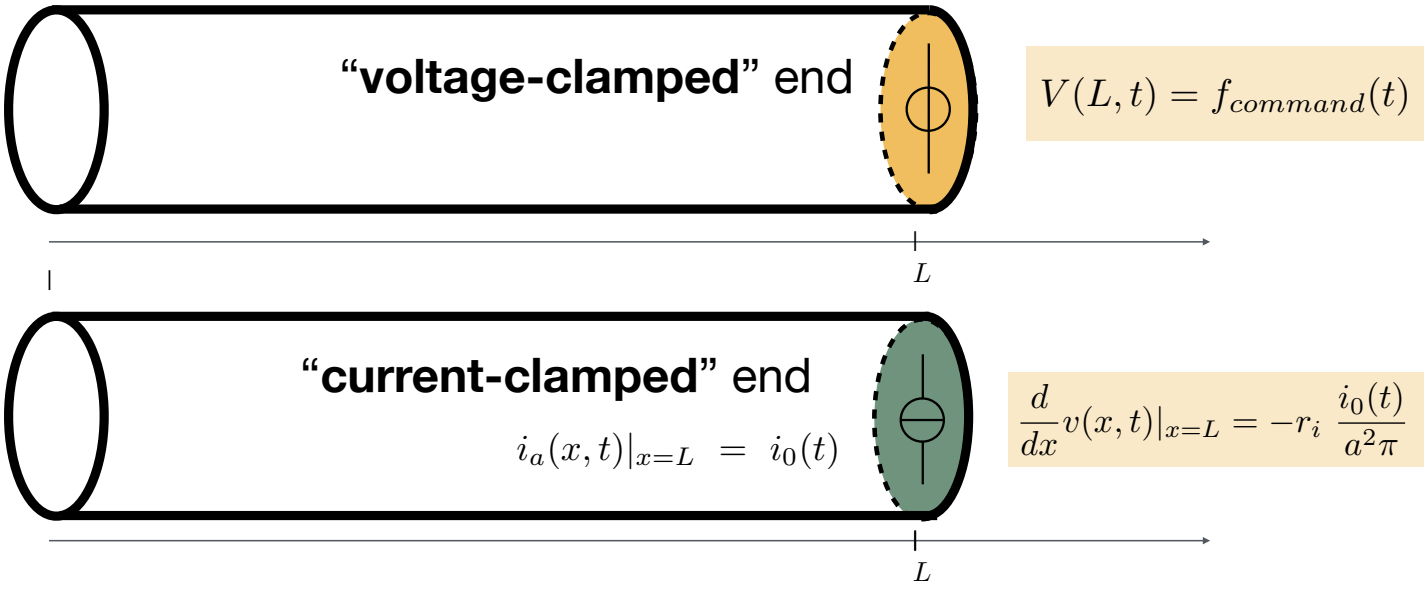


$$-\frac{1}{r_i}\frac{d}{dx}v(x,t)|_{x=L} = \frac{1}{R_0\ a^2\pi}V(L,t)$$

L

"load-attached" in electrical *jargon*!!

Boundary conditions 4



Let's solve that *PDE!*

- steady-state regimes

- transient regimes

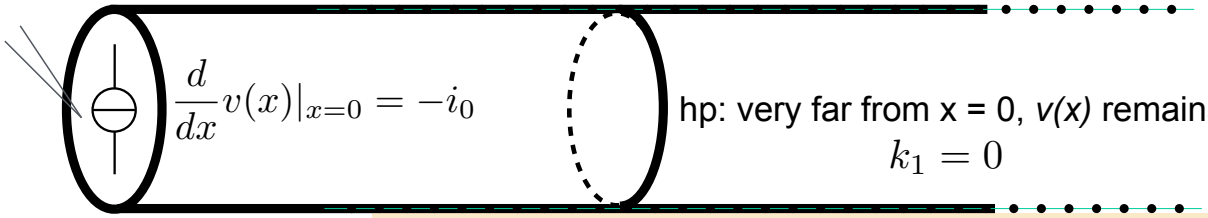
Steady-state (w.r.t. time) - DC regime
semi-**infinite** cable…

$$\frac{\partial v(x,t)}{\partial t} = 0 \qquad v(x,t) \to v(x)$$

$$\lambda^2 \frac{\partial^2 v(x,t)}{\partial x^2} = \tau_m \frac{\partial v(x,t)}{\partial t} + v(x,t) \qquad \lambda^2 \frac{d^2 v(x)}{dx^2} = v(x) \qquad \lambda^2 s^2 = 1$$

$$v(x) = k_1 e^{x/\lambda} + k_2 e^{-x/\lambda}$$

$$s_1 = +\frac{1}{\lambda} \qquad s_2 = -\frac{1}{\lambda}$$



$$\frac{d}{dx} v(x)|_{x=0} = -i_0$$

hp: very far from x = 0, *v(x)* remains <u>finite</u>

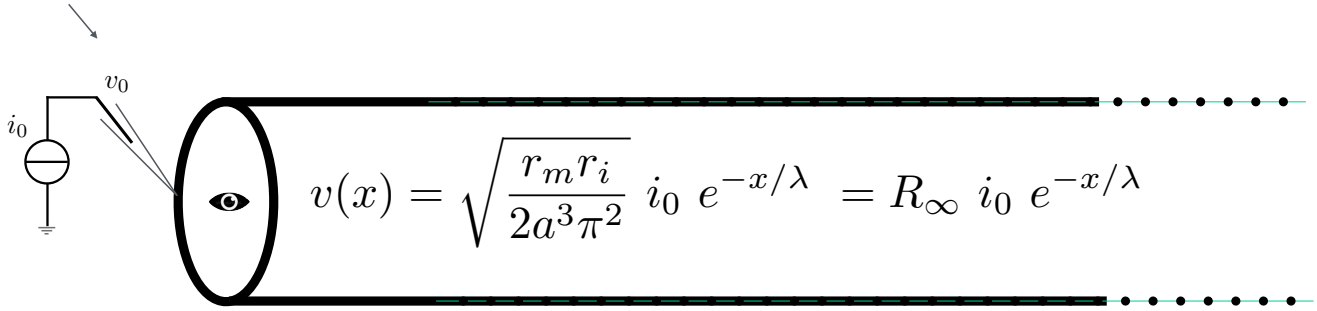
$$k_1 = 0$$

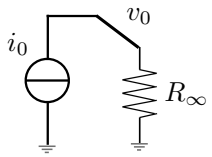
$$-\frac{a^2 \pi}{r_i \lambda} k_2 = i_0 \qquad k_2 = \sqrt{\frac{r_m r_i}{2a^3 \pi^2}}\ i_0$$

$$v(x) = \sqrt{\frac{r_m r_i}{2a^3 \pi^2}}\ i_0\ e^{-x/\lambda} = R_\infty\ i_0\ e^{-x/\lambda}$$

input resistance

Steady-state (w.r.t. time) - DC regime
equivalent (Thevenin) circuit





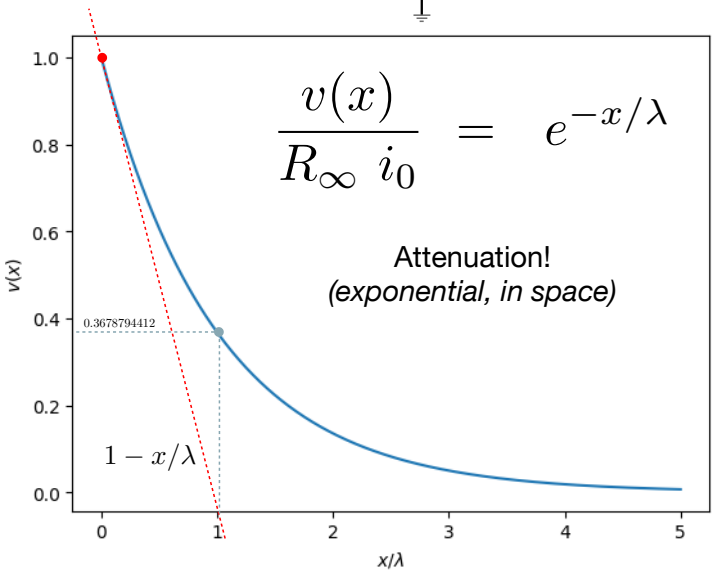
I *see* just an <u>equivalent</u> resistor/conductance
from this point of the “circuit”…

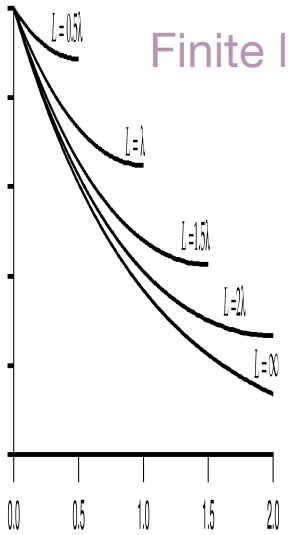
$$v_0 = R_\infty\ i_0$$

Steady-state (i.e. time) - semi-**infinite** cable

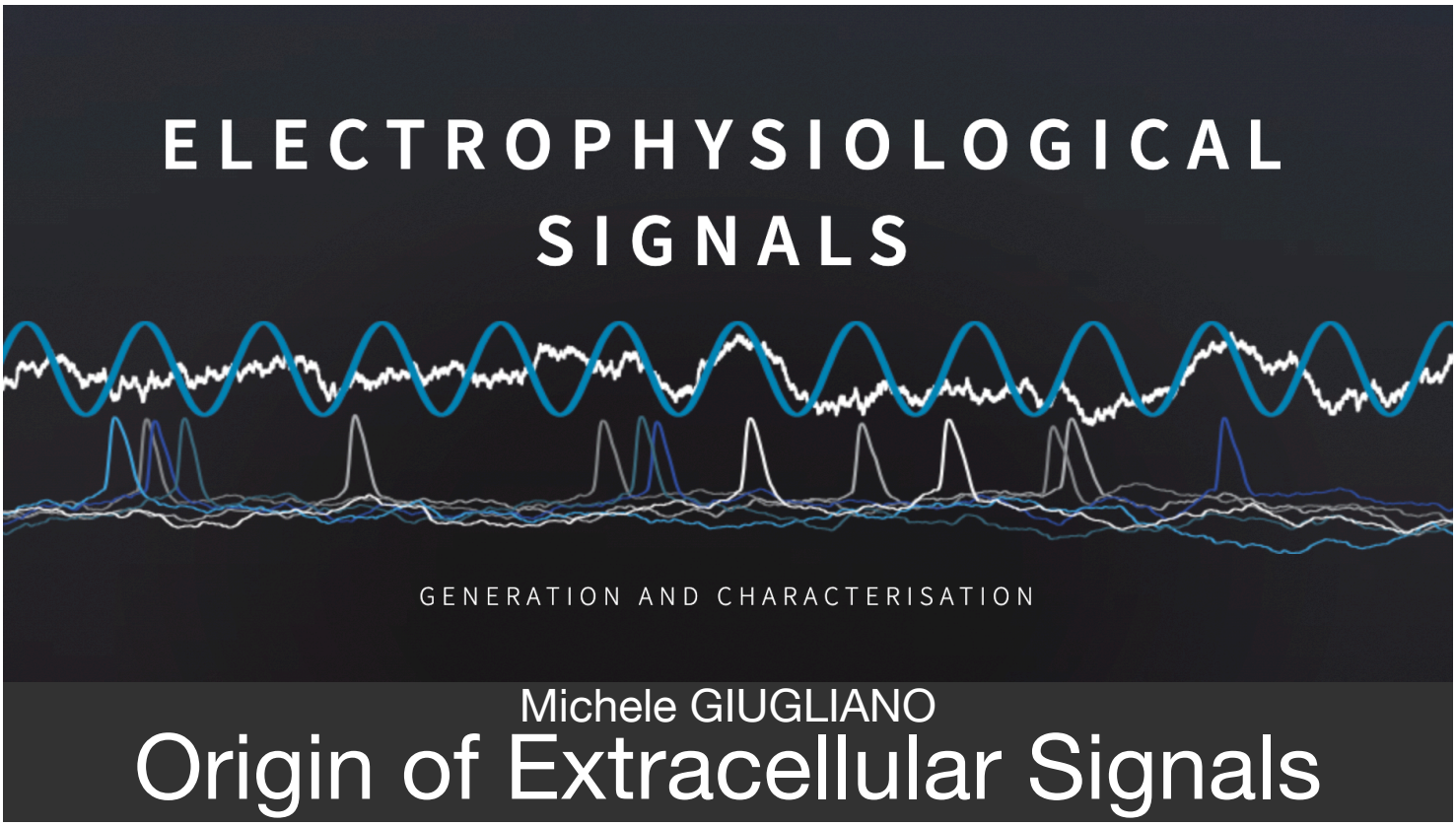
$$a = 2\mu m$$
$$\lambda = 1000\mu m$$

$$v(x) = R_\infty \; i_0 \; e^{-x/\lambda}$$





Finite length!



Michele GIUGLIANO

Origin of Extracellular Signals

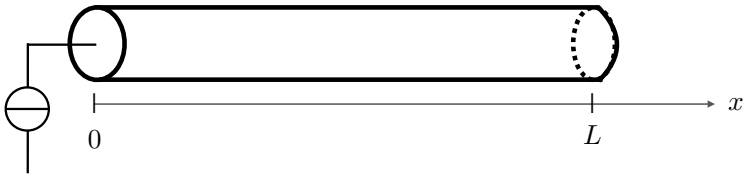
ATTENDANCE TRACKING - **code T3LXY**

(for statistical purposes only)

https://www.unimore.it/it/servizi/unimore-app

Steady-state (i.e. time) - **finite** cable

$\frac{d}{dx}v(x,t)|_{x=0} = -r_i \frac{i_0(t)}{\boldsymbol{a}^2\pi}$



$\frac{d}{dx}v(x,t)|_{x=L} = 0$

$$v(x) = k_1 e^{x/\lambda} + k_2 e^{-x/\lambda} \qquad v(x) = \frac{\hat{a}+\hat{b}}{2} e^{x/\lambda} + \frac{\hat{a}-\hat{b}}{2} e^{-x/\lambda}$$

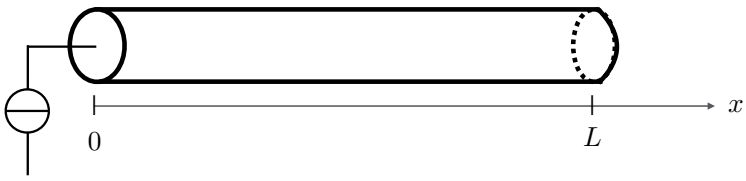
$$v(x) = \frac{\hat{a}}{2}\left(e^{x/\lambda} + e^{-x/\lambda}\right) + \frac{\hat{b}}{2}\left(e^{x/\lambda} - e^{-x/\lambda}\right)$$

$sinh(x) = \frac{e^x - e^{-x}}{2}$ $\quad cosh(x) = \frac{e^x + e^{-x}}{2}$ $\quad \frac{d}{dx}\cosh(x) = \sinh(x)$ $\quad \frac{d}{dx}\sinh(x) = \cosh(x)$

$$v(x) = \hat{a}\ \cosh(x/\lambda) + \hat{b}\ \sinh(x/\lambda)$$
$$\frac{d}{dx}v(x) = \frac{\hat{a}}{\lambda}\sinh(x/\lambda) + \frac{\hat{b}}{\lambda}\cosh(x/\lambda)$$

Steady-state (i.e. time) - **finite** cable

$\frac{d}{dx}v(x,t)|_{x=0} = -r_i \frac{i_0(t)}{\boldsymbol{a}^2\pi}$



$\frac{d}{dx}v(x,t)|_{x=L} = 0$

$$v(x) = \hat{a}\ \cosh(x/\lambda) + b\ \sinh(x/\lambda)$$
$$\frac{d}{dx}v(x) = \frac{\hat{a}}{\lambda}\sinh(x/\lambda) + \frac{\hat{b}}{\lambda}\cosh(x/\lambda)$$

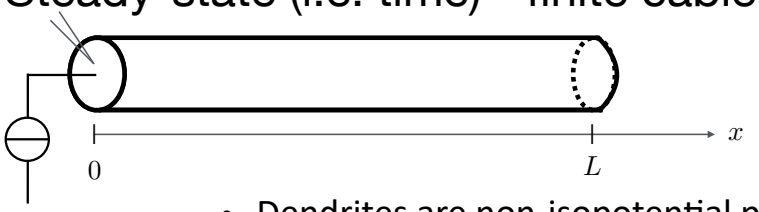
$\frac{\hat{b}}{\lambda} = -r_i \frac{i_0}{\boldsymbol{a}^2\pi}$ $\quad \hat{a}\ sinh(L/\lambda) + \hat{b}\ cosh(L/\lambda) = 0$ $\quad \hat{a} = -\hat{b}\ \frac{cosh(L/\lambda)}{sinh(L/\lambda)}$

$$v(x) = -\hat{b}\ \frac{cosh(L/\lambda)cosh(x/\lambda) - sinh(L/\lambda)sinh(x/\lambda)}{sinh(L/\lambda)}$$

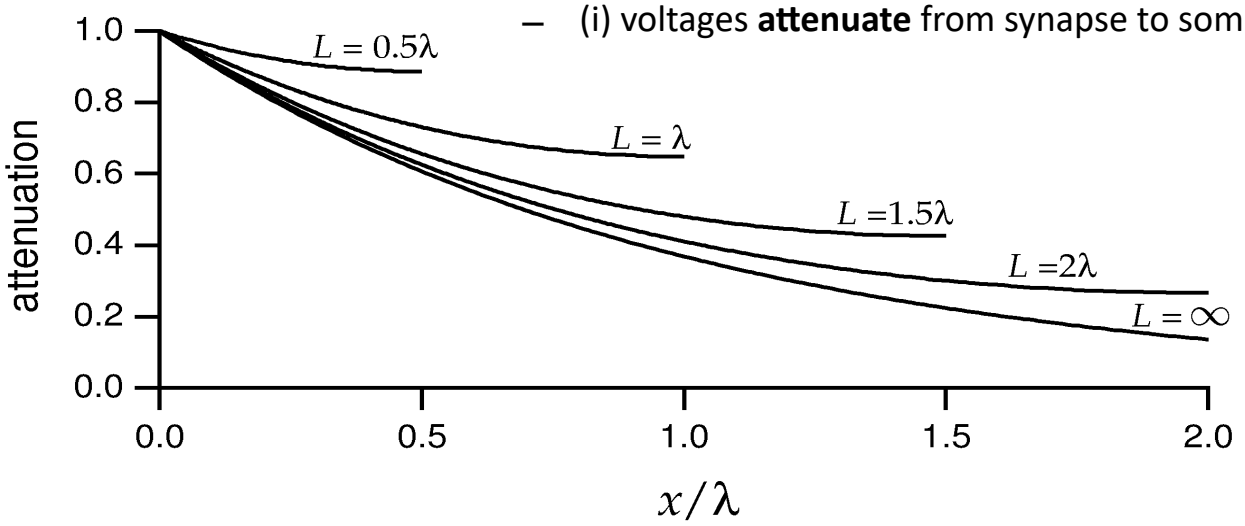
$$v(x) = r_i\ \lambda \frac{i_0}{a^2\pi}\ \frac{cosh[(L-x)/\lambda]}{sinh(L/\lambda)}$$

Steady-state (i.e. time) - finite cable

$a = 2\mu m$
$\lambda = 1000\mu m$

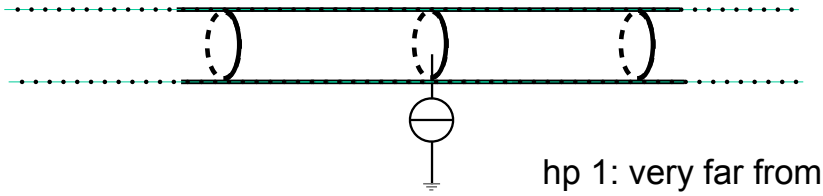


- Dendrites are non-isopotential portions of a neuron!
  - (i) voltages **attenuate** from synapse to soma!



Steady-state (i.e. time) - infinite cable

$a = 2\mu m$
$\lambda = 1000\mu m$



hp 1: very far from x = 0, *v(x)* remains finite
hp 2: continuity of *v(x)* for x = 0
hp 3: slightly different boundary condition

$v(x) = k_1 e^{x/\lambda} + k_2 e^{-x/\lambda}$ $\quad v(x) = k\ e^{-|x|/\lambda}$

$$k = \frac{1}{2}\sqrt{\frac{r_m\ r_i}{2\pi^2 a^3}}\ I_0$$

$$v(x) = \frac{1}{2} R_\infty e^{-|x|/\lambda}$$

$$\lambda^2 \frac{\partial^2 V(x,t)}{\partial x^2} = \tau_m \frac{\partial V(x,t)}{\partial t} + (V(x,t) - E) + r_m \; [g_{syn}(x,t)(V(x,t) - E_{syn}) - i_{ext}(x,t))]$$

$$I_{ext}(x,t) \; = \; I_0 \; \Delta x \; \delta(x) \qquad i_{ext} = \frac{I_{ext}(x,t)}{2\pi a \Delta x} \qquad \lambda = \sqrt{\frac{a}{2}\frac{r_m}{r_i}}$$

$$\lambda^2 \frac{\partial^2 V(x)}{\partial x^2} = 0 + (V(x,t) - E) \; - \; r_m \; \frac{I_0}{2\pi a} \delta(x)$$

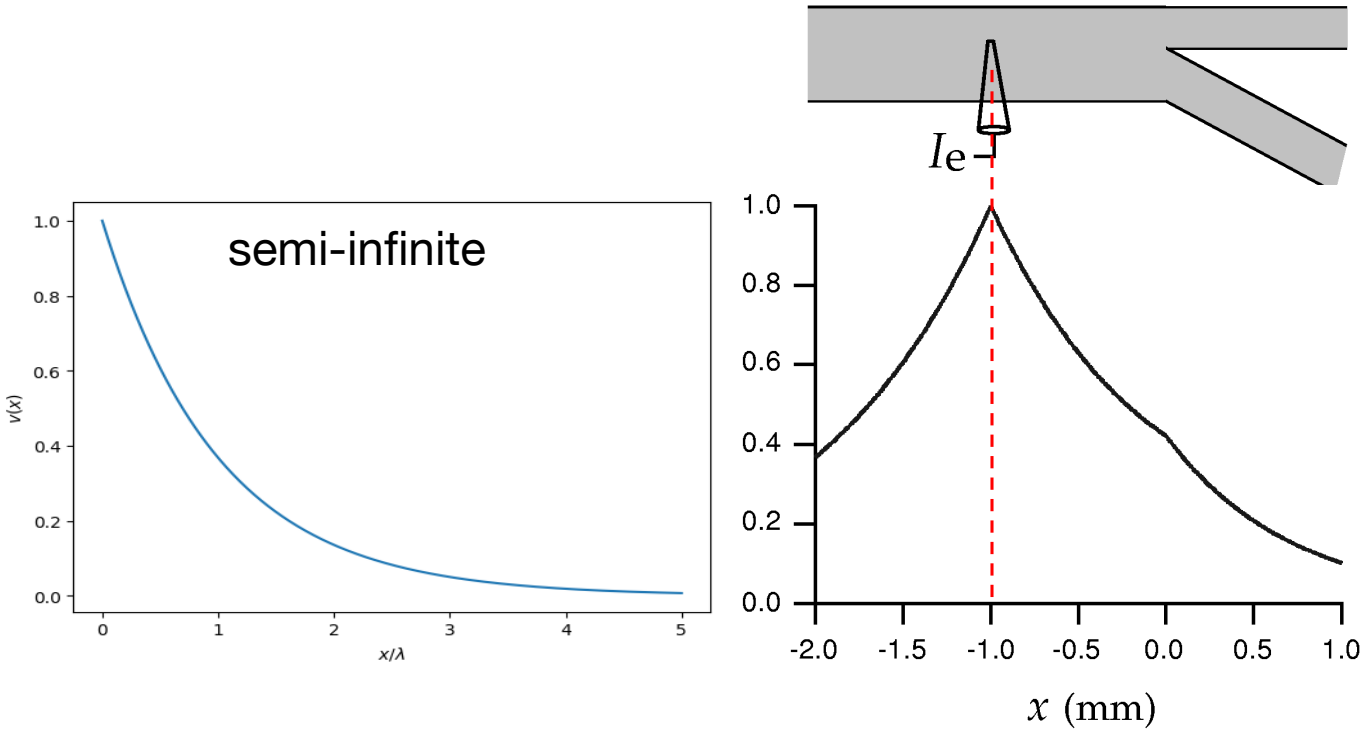
$$\int_{0-}^{0+} \lambda^2 \frac{\partial^2 V(x)}{\partial x^2} dx = \int_{0-}^{0+} (V(x,t) - E) dx \; - \; r_m \; \frac{I_0}{2\pi a} \int_{0-}^{0+} \delta(x) dx$$

$$\lambda^2 \frac{dV(x)}{dx}\Big|_{0-}^{0+} = 0 - \; r_m \; \frac{I_0}{2\pi a} \qquad V(x) = E + k \; e^{-|x|/\lambda}$$

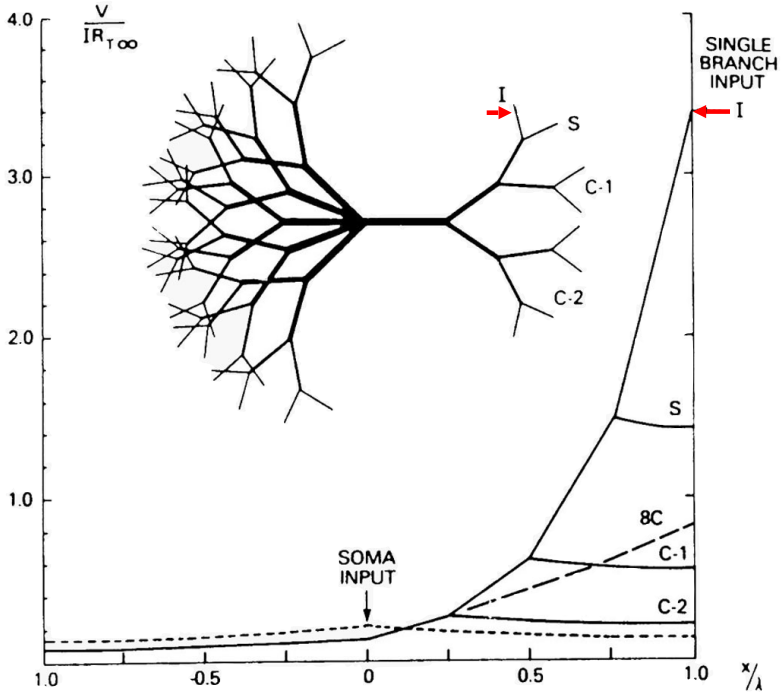
$$\frac{dV(x)}{dx}\Big|_{0-}^{0+} = -\frac{k}{\lambda} \; - \; (+\frac{k}{\lambda}) \qquad k \; = \; \frac{r_m}{2\lambda \; 2\pi a} \; I_0$$

$$k \; = \; \frac{1}{2}\sqrt{\frac{r_m \; r_i}{2\pi^2 a^3}} \; I_0 \qquad V(x) = E + \; \frac{1}{2}\sqrt{\frac{r_m \; r_i}{2\pi^2 a^3}} \; I_0 \; e^{-|x|/\lambda} \; = \; E + \; \frac{1}{2} R_\infty e^{-|x|/\lambda}$$

Steady-state (i.e. time) - infinite cable

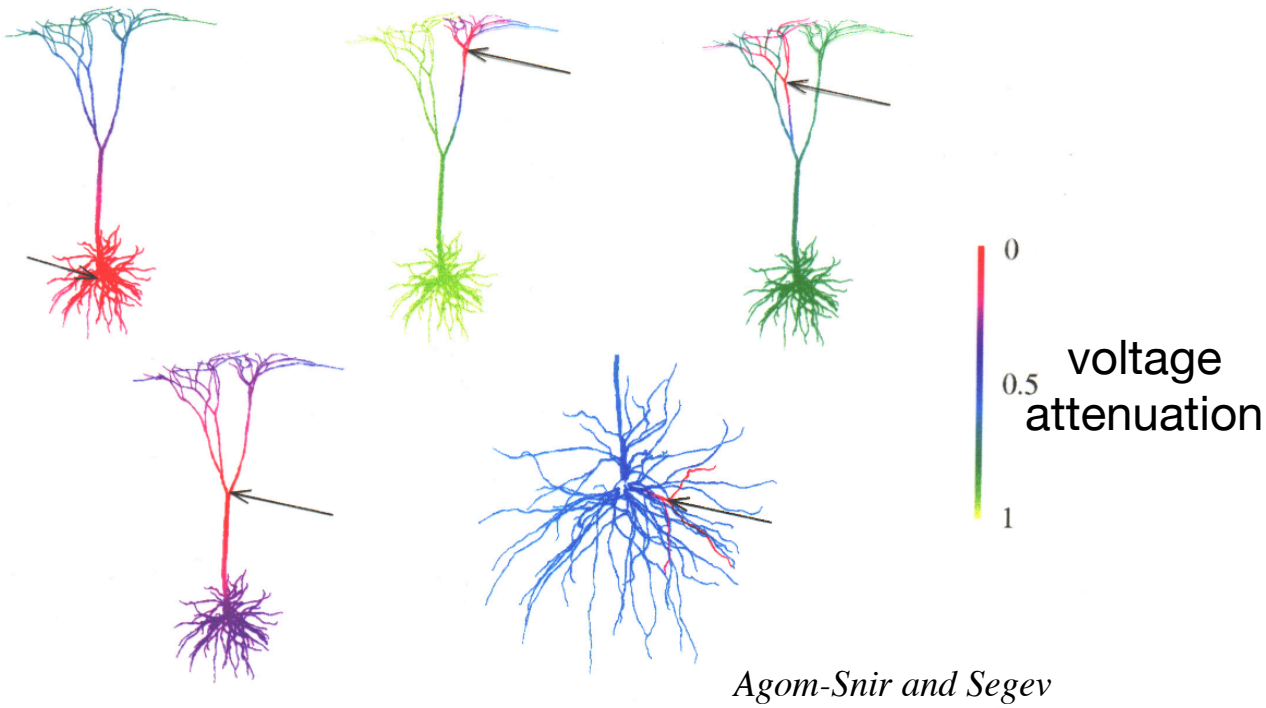


What do we learn? 1) Steep & asymmetrical voltage attenuation (from dendrites to the soma); 2) existence of synaptic *territories*



Rall & Rinzel, 1973

Dendritic “functional subunits” (“synaptic territory”)



Agom-Snir and Segev

What about…

transients?

Sinusoidal Regime - AC regime
semi-infinite cable…

$$\lambda^2 \frac{\partial^2 v(x,t)}{\partial x^2} = \tau_m \frac{\partial v(x,t)}{\partial t} + v(x,t)$$

$$W(x,\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} v(x,t) e^{-j\omega t} dt$$

Fourier transform: *time* “disappears”…

$$\lambda^2 \frac{d^2 W(x,\omega)}{dx^2} = \tau_m \; j \; \omega \; W(x,\omega) + W(x,\omega)$$

$$\frac{\lambda^2}{1 + \tau_m \; j \; \omega} \frac{d^2 W(x,\omega)}{dx^2} = W(x,\omega)$$

$$W(x,\omega) = R_\infty \; I_0(\omega) \; e^{-x/\lambda^*} \qquad \lambda^* = \frac{\lambda}{\sqrt{1 + \tau_m j\omega}}$$

$$W(x,\omega) = R_\infty \; I_0(\omega) \; M \; e^{j\Phi} \qquad M = M(x,\omega) = ?$$

e^{-x/λ^*} Before that… I do have a Frequency-dependent and complex space-constant

$$\frac{1}{\lambda^*} = \frac{\sqrt{1+\tau_m j\omega}}{\lambda} = A\ e^{j\ \phi} \qquad A = \frac{\sqrt[4]{1+\tau_m^2\omega^2}}{\lambda}$$

$$\phi = \frac{1}{2}\arctan \tau_m\omega$$

$$e^{j\ \phi} = cos(\phi) + j\ sin(\phi)$$

Euler's formula
the most beautiful equation in mathematics

$$e^{-x/\lambda^*} = e^{-x\ Acos(\phi)\ -j\ xAsin(\phi)}$$

$$cos(\arctan k) = \frac{1}{\sqrt{1+k^2}} \qquad cos(\alpha/2) = \pm\sqrt{\frac{1+cos(\alpha)}{2}}$$

$$sin(\arctan k) = \frac{k}{\sqrt{1+k^2}} \qquad sin(\alpha/2) = \pm\sqrt{\frac{1-cos(\alpha)}{2}}$$

$$cos(0.5\arctan k) = \pm\sqrt{\frac{1+\frac{1}{\sqrt{1+k^2}}}{2}}$$

$$W(x,\omega) = R_\infty\ I_0(\omega)\ M\ e^{j\Phi} \qquad M = M(x,\omega) = ?$$

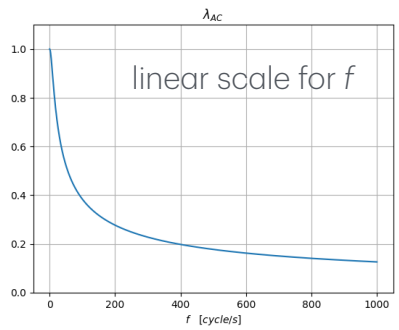
$$A = \frac{\sqrt[4]{1+\tau_m^2\omega^2}}{\lambda} \qquad \phi = \frac{1}{2}\arctan \tau_m\omega \qquad e^{j\ \phi} = cos(\phi) + j\ sin(\phi)$$

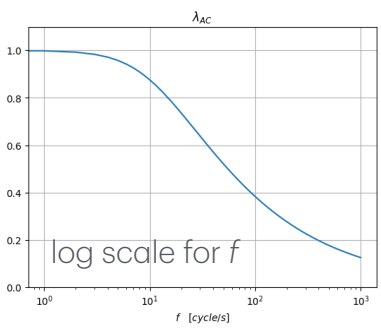
$$W(x,\omega) = R_\infty\ I_0(\omega)\ e^{-xAcos(\phi)}\ e^{-jxAsin(\phi)}$$

DC

$$v(x) = \sqrt{\frac{r_m r_i}{2a^3\pi^2}}\ i_0\ e^{-x/\lambda} = R_\infty\ i_0\ e^{-x/\lambda}$$

$$e^{-x/\lambda_{AC}} \qquad \lambda_{AC} = \frac{1}{Acos(\phi)} = \lambda\sqrt{\frac{2}{1+\sqrt{1+(2\pi f)^2\tau_m^2}}}$$

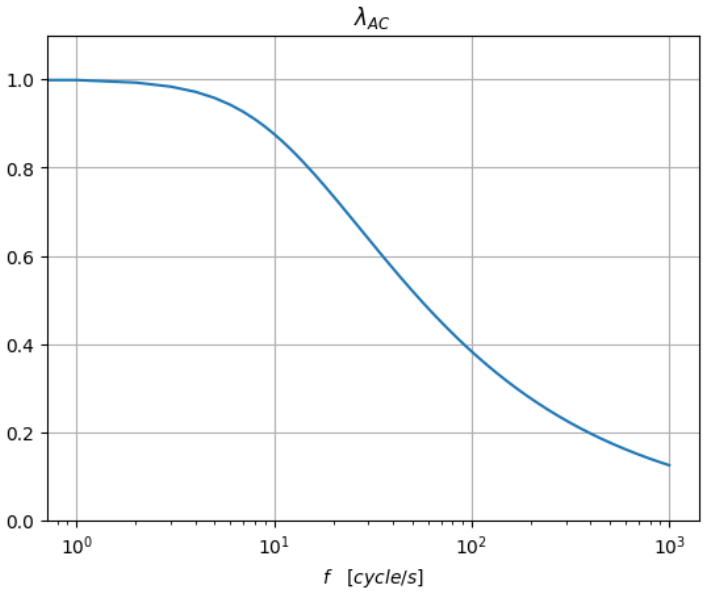




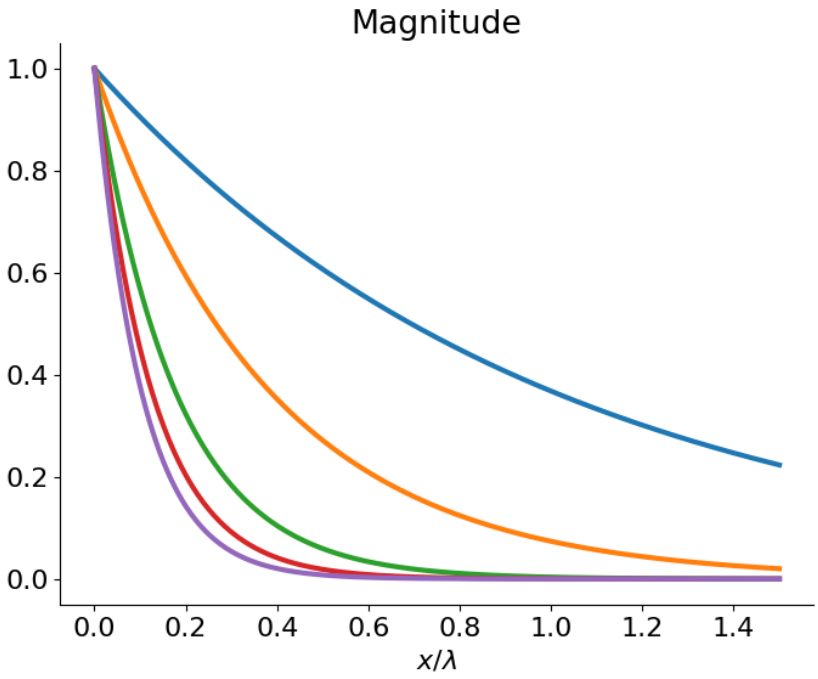
hp: 20ms membrane time constant

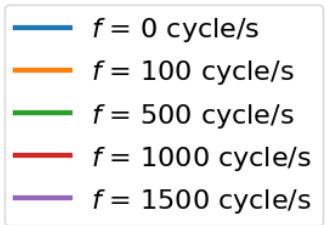
Sinusoidal Regime - AC - attenuation (semi-infinite cable…)

- Dendrites are “filtering out” fast frequency components. (i) The (effective, equivalent) space-constant *shrinks* for fast frequencies!



Sinusoidal Regime - AC - attenuation (semi-infinite cable…)





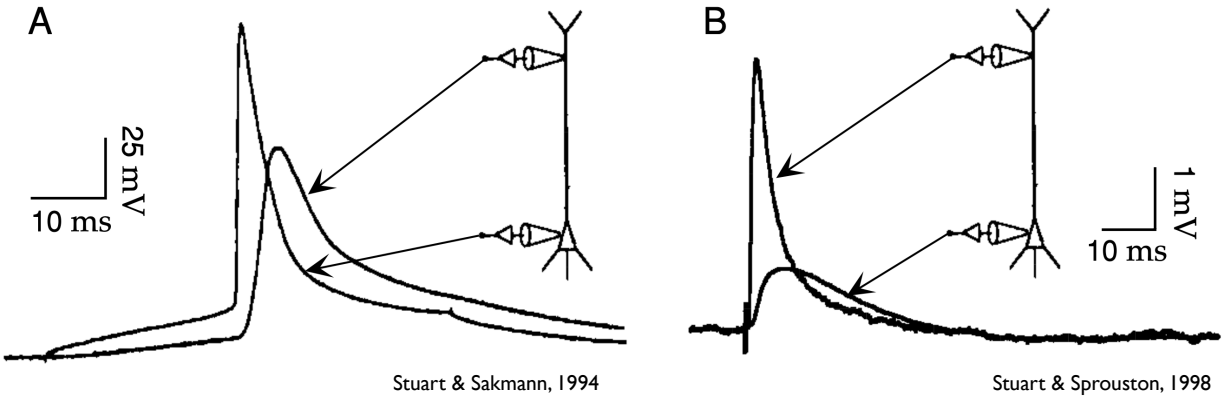
$$e^{-x/\lambda_{AC}}$$

$$\lambda_{AC} = \lambda \sqrt{\frac{2}{1+\sqrt{1+(2\pi f)^2 \tau_m^2}}}$$

hp: 20ms membrane time constant

Of cables: synapses, dendrites, and somata (and back)

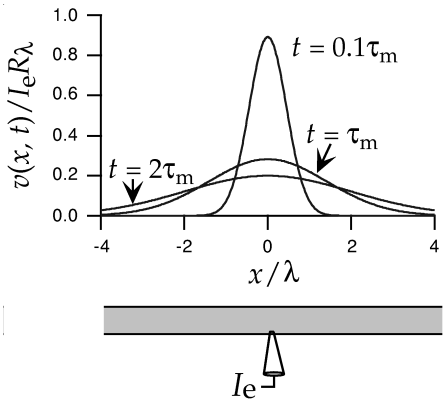
Back-propagating AP and (forward propagating) EPSPs



Stuart & Sakmann, 1994

Stuart & Sprouston, 1998

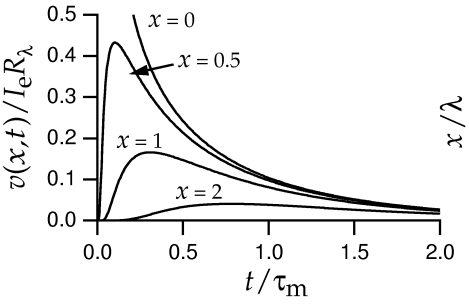
Transient solution (attenuation, shape change, delay)

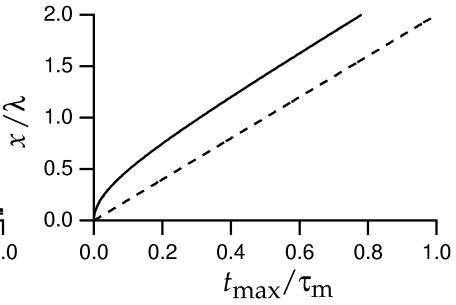


$$V(x,t) = \frac{I_e\, R_\lambda}{\sqrt{4\pi\, \lambda_m^2\, t/\tau_m}} \exp\left(-\frac{\tau_m\, x^2}{4\lambda_m^2\, t}\right) \exp\left(-\frac{t}{\tau_m}\right)$$

$$V(x,t) = \frac{A(t)}{\sqrt{2\pi\, \sigma^2(t)}} \exp\left(-\frac{x^2}{2\sigma^2(t)}\right)$$

$$A(t) = I_e\, R_\lambda \exp\left(-\frac{t}{\tau_m}\right), \qquad \sigma(t) = \lambda_m \sqrt{\frac{2\,t}{\tau_m}} \qquad R_{\lambda_i} = \frac{r_L\, \lambda_i}{\pi\, a_i^2}$$

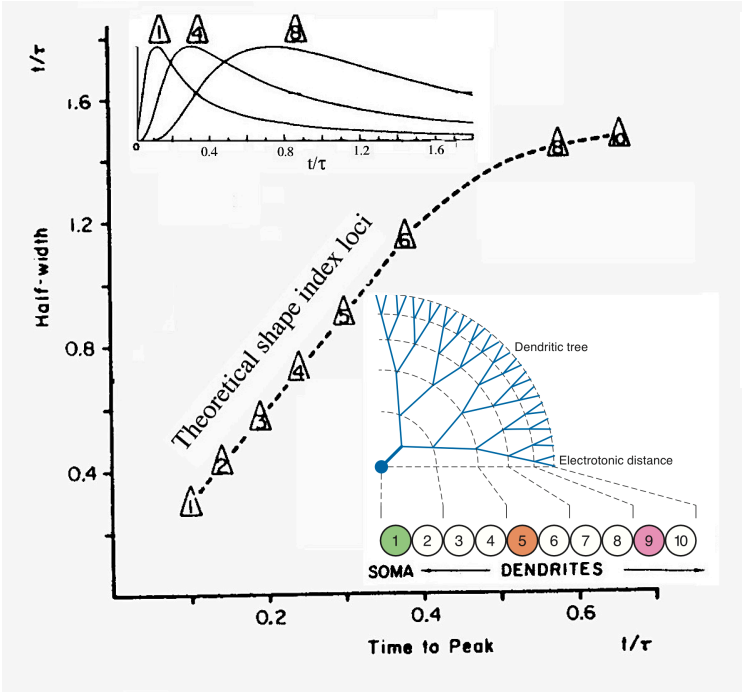


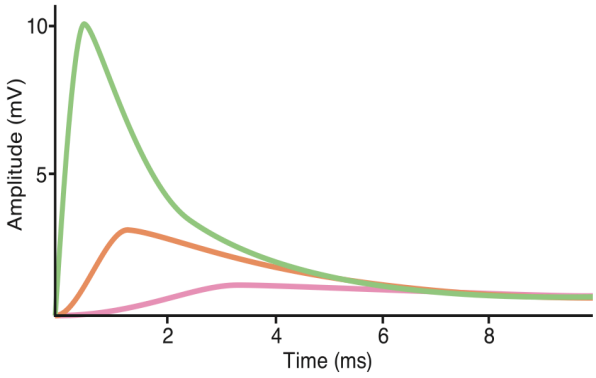


$$t_{max} \approx \frac{x\, \tau_m}{2\, \lambda_m} \quad \Rightarrow$$

$$\Rightarrow \quad v_{cond} \equiv \frac{x}{t_{max}} \approx 2\,\frac{\lambda_m}{\tau_m}$$

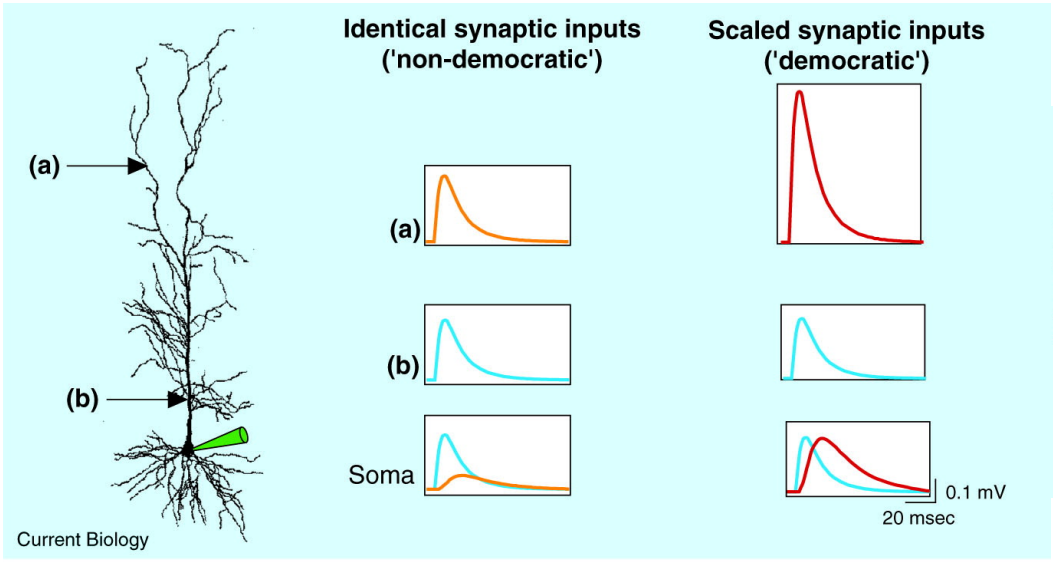
Experimental Predictions
distal synapses are “broader” and “delayed”

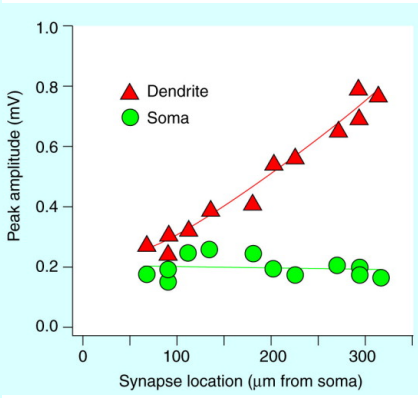




Synaptic potentials attenuate from their origin towards other regions of the dendrites/neuron

Dendritic “democracy” (in CA1 pyramidal neurons)

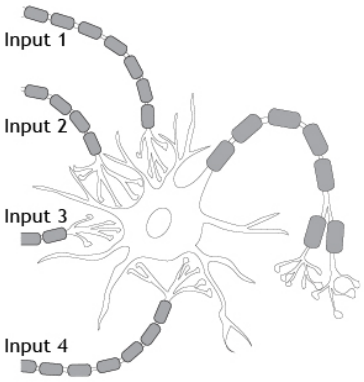


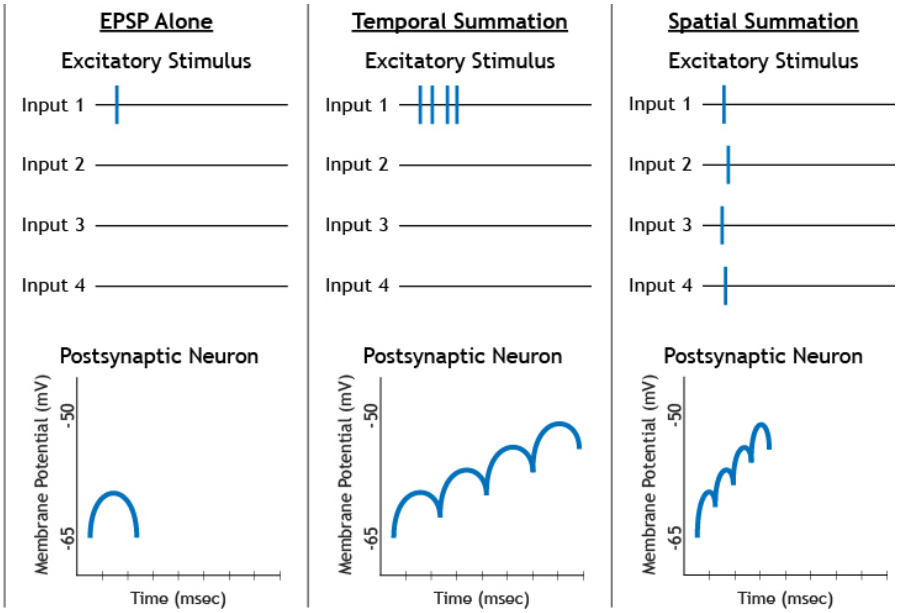


CA1 pyramidal neuron

Häusser, 2001

Temporal and Spatial summation





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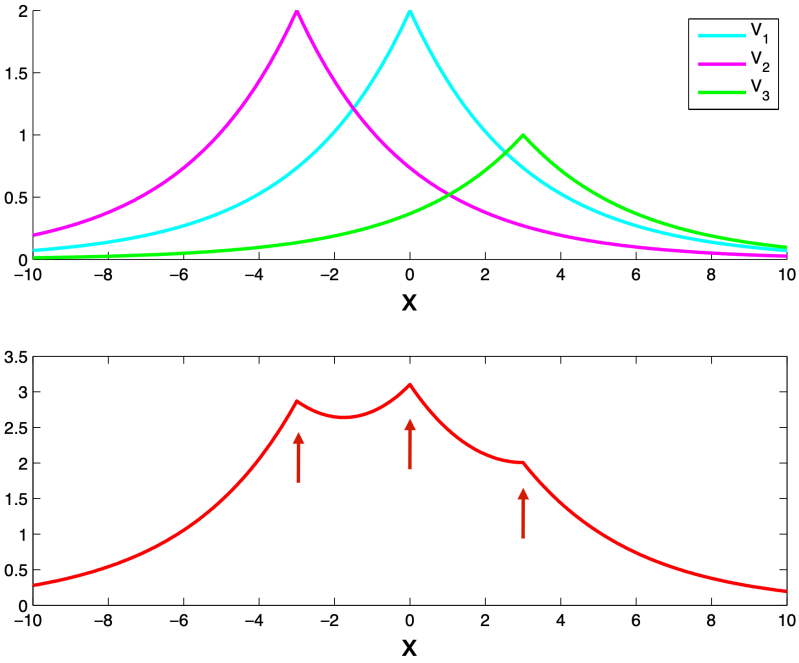
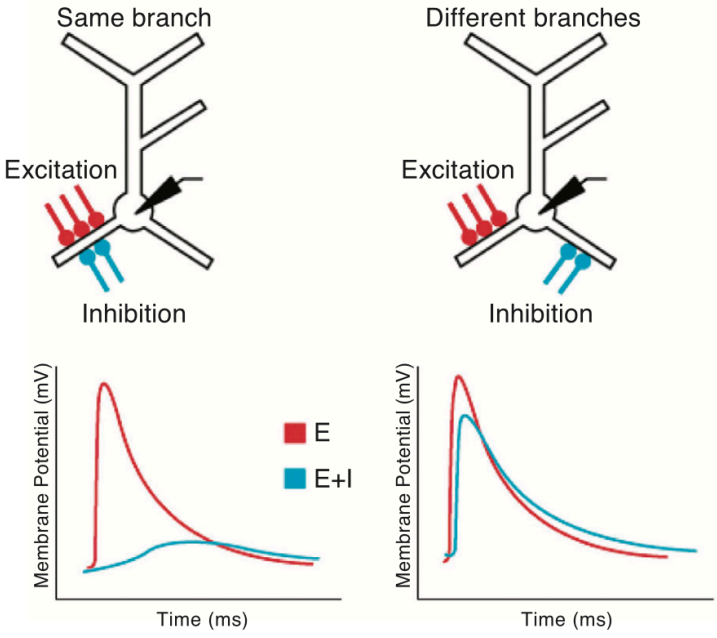
Temporal and (linear) Spatial summation

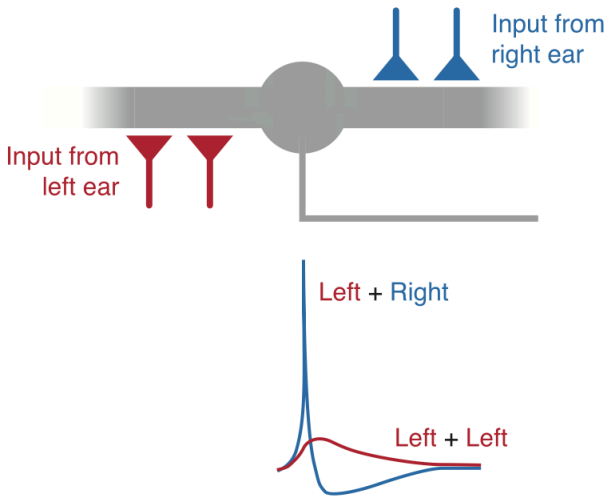
$I_{sum}(x) = I_1(x) + I_2(x) + I_3(x)$

$V_{sum} = V_1(x) + V_2(x) + V_3(x)$



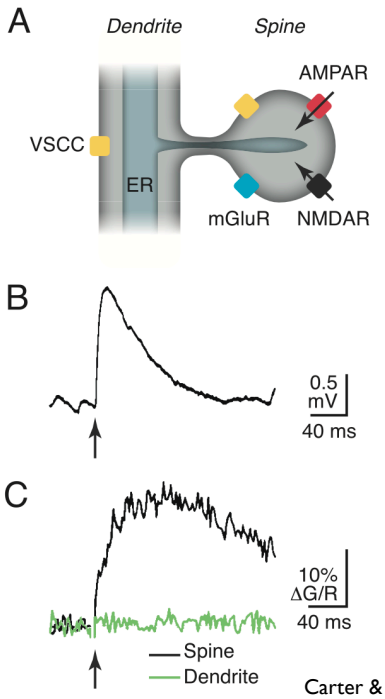
(sub)linear Spatial Summation



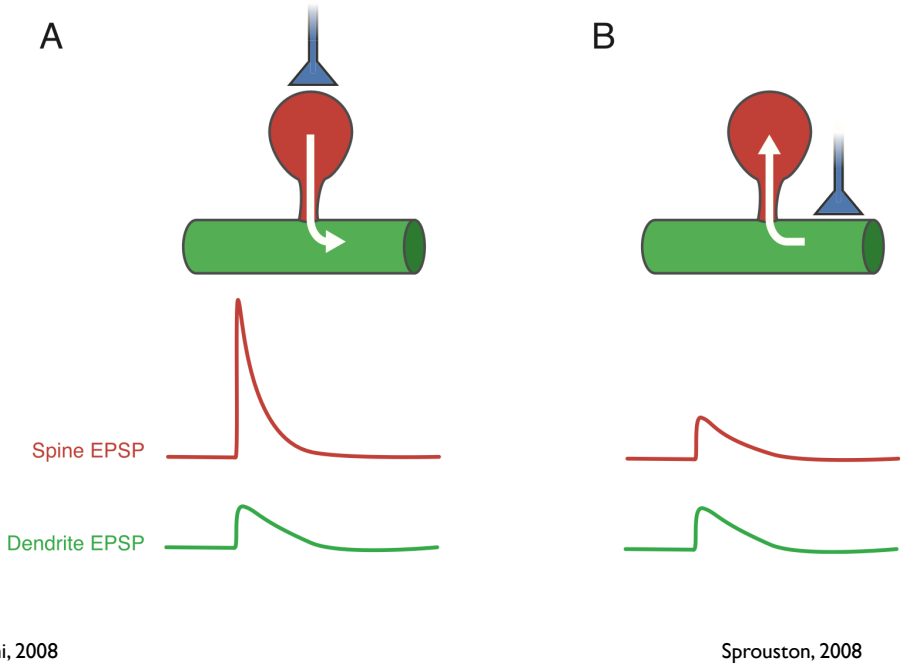


London & Hausser, 2005

Dendritic spines, depolarization, and Ca++ influx

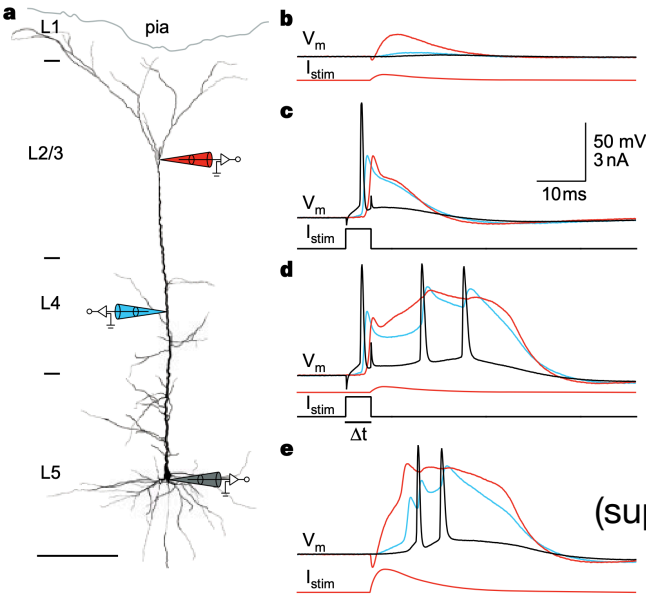


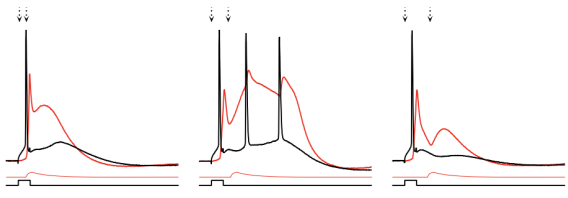
Carter & Sabatini, 2008



Sprouston, 2008

This **b**ack-propagating **A**P activated **C**a2+ spike firing (BAC) is a coincidence detection mechanism (distal inputs & AP firing)

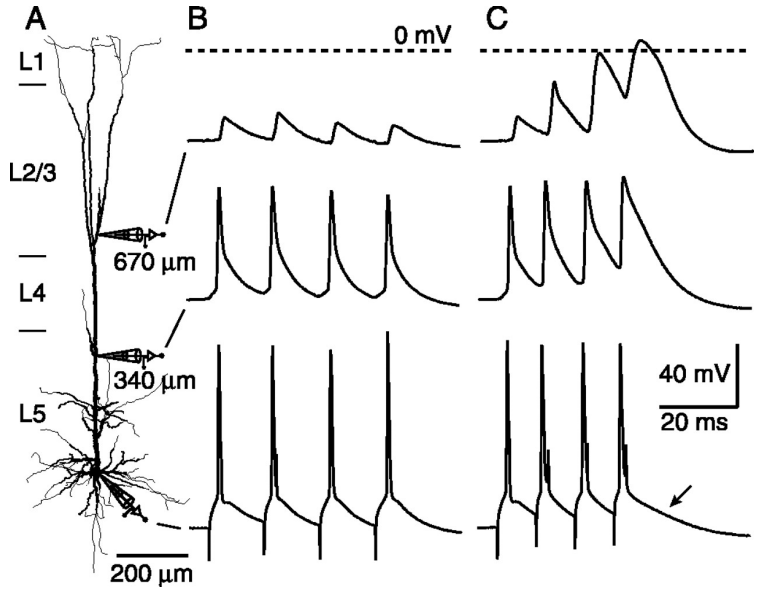




(supra)-linear Spatial Summation

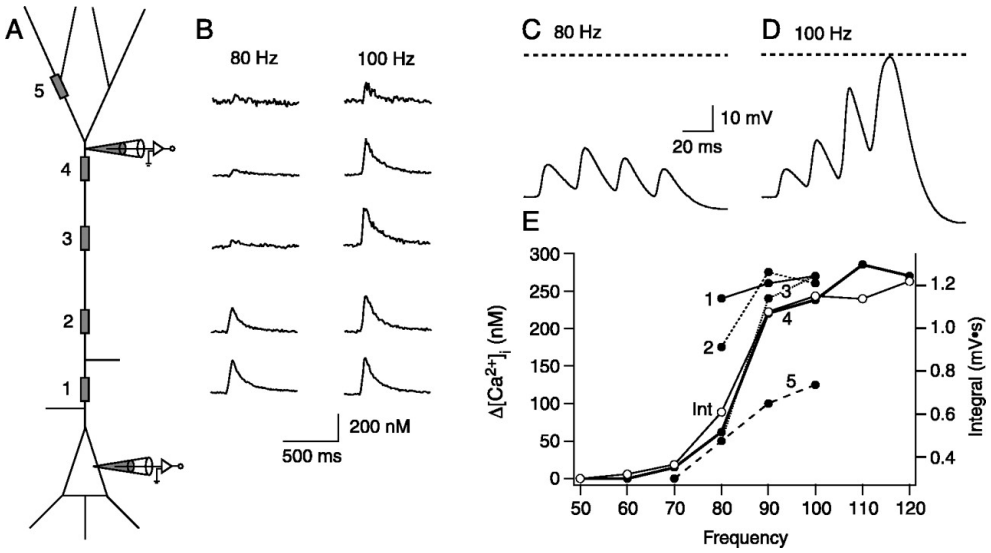
Larkum, Zhu, & Sakmann, 1999

Ca++ electrogenesis in distal apical dendrites of L5 pyr. cells at a critical frequency of back-propagating AP



Larkum et al. 1999

Ca++ *electrogenesis* in distal apical dendrites of L5 pyr. cells at a critical frequency of back-propagating AP



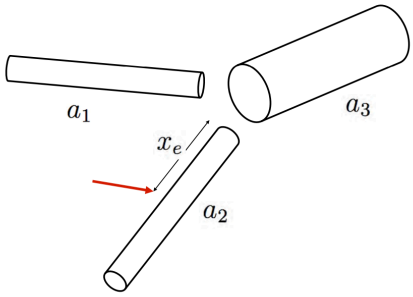
Larkum et al. 1999

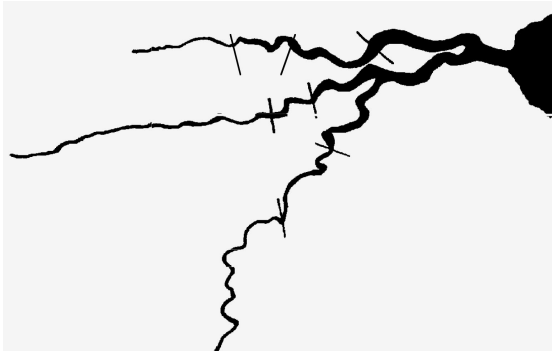
Branching points

$$v_1(x) = I_e R_{\lambda_1} p_1 \exp\left(-\frac{x}{\lambda_1} - \frac{x_e}{\lambda_2}\right)$$

$$v_2(x) = I_e R_{\lambda_2} \left[(p_2 - 1) \exp\left(-\frac{x}{\lambda_2} - \frac{x_e}{\lambda_2}\right) + \exp\left(-\frac{|x_e - x|}{\lambda_2}\right)\right]$$

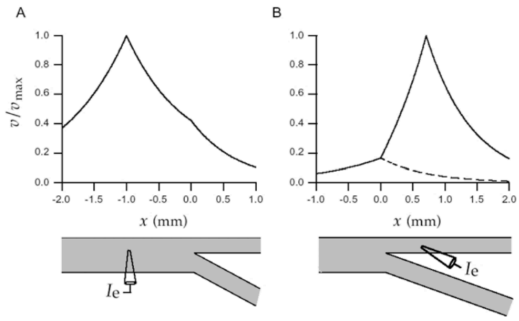
$$v_3(x) = I_e R_{\lambda_3} p_3 \exp\left(-\frac{x}{\lambda_3} - \frac{x_e}{\lambda_2}\right)$$





thick → thin

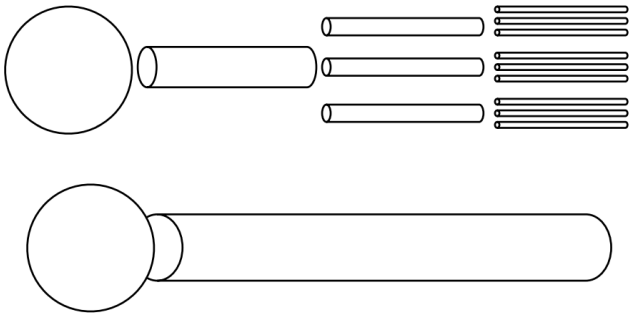
thin → thick



$$p_i = \frac{a_i^{3/2}}{a_1^{3/2} + a_2^{3/2} + a_3^{3/2}}$$

$$R_{\lambda_i} = \frac{r_L \lambda_i}{\pi a_i^2}$$

Rall's model



Radius a and length L of the equivalent dendrite are chosen to match the properties of the actual dendritic tree:
(i) Surface area $2\pi L a$ is set equal to surface area of full dendritic tree.
(ii) Electrotonic length L/λ is set equal to total electrotonic length of tree segments $\Sigma_i \frac{L_i}{\lambda_i}$

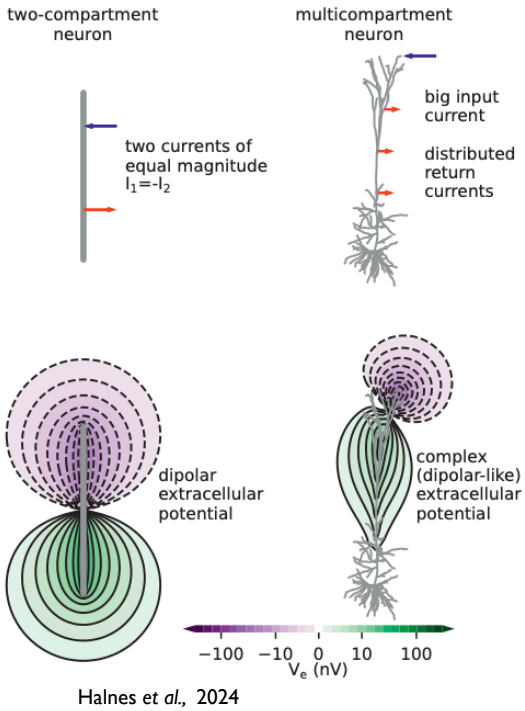
The Rall model is exact when

$$a_1^{3/2} = a_2^{3/2} + a_3^{3/2} \quad \Rightarrow \quad p_1 = p_2 + p_3 = \frac{1}{2}$$

Let’s discuss extracellular signals and volume-conductor theory

Let’s focus on space-dependent electrical properties and electrical phenomena of neurons. (cable theory)

Extracellular signals: more in general, how to?



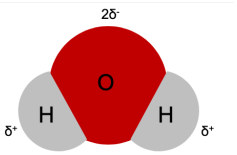
Halnes *et al.*, 2024

- Describing, **across space**, ionic currents changes. (**cable theory**)

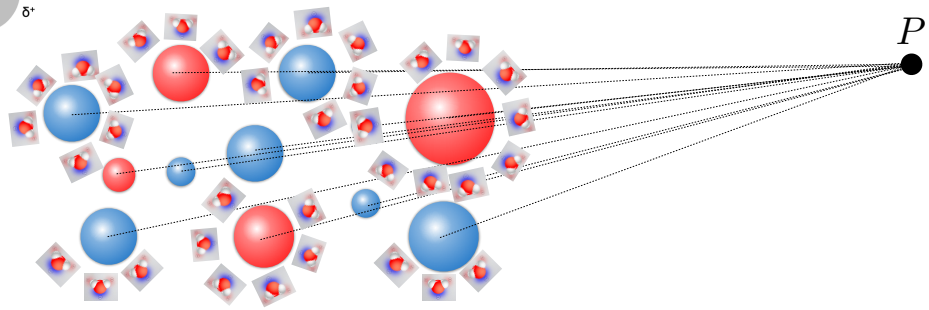
$$\lambda^2 \frac{\partial^2 v(x,t)}{\partial x^2} = \tau_m \frac{\partial v(x,t)}{\partial t} + v(x,t)$$

- Computing extracellular potentials, from the **spatial distribution of transmembrane currents**. (**volume-conductor theory**)

$$V_e(\mathbf{r}) = \sum_{\mathbf{n}} \frac{\mathbf{i_n}}{4\pi\sigma_\mathbf{t}|\mathbf{r}-\mathbf{r_n}|}$$



*Extra*cellular potential *Ve*



$r_D \approx 1nm$

Debye shielding distance

$$V_{total}(P) = \frac{1}{4\pi\epsilon_r\epsilon_0}\left(\frac{Q_1 e^{-r_1/r_D}}{r_{P-Q_1}} + \frac{Q_2 e^{-r_2/r_D}}{r_{P-Q_2}} + ... + \frac{Q_M e^{-r_M/r_D}}{r_{P-Q_M}}\right)$$

Very hard to use for macroscopic distances, where *shielding* in space and ultrafast *relaxation* times must be considered.

(Across the plasma membrane was ok to use it!)

*Extra*cellular potential *Ve*

⇒ Particles move with a velocity proportional (by u - *mobility*) to the external force field

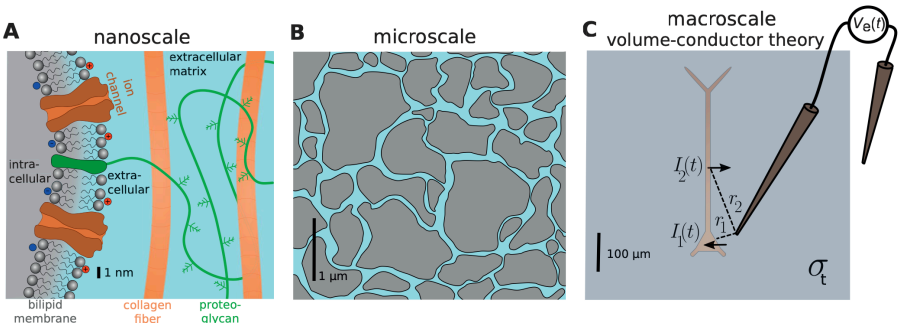
$$F_{ext} + F_{friction} = m\frac{dv(t)}{dt} \qquad F_{friction} = -\lambda\, v(t)$$

$$v(t) = k\, e^{-\frac{\lambda}{m}t} + \frac{F_{ext}}{\lambda}$$

$$\frac{\lambda}{m} >> 1$$

$$\frac{m}{\lambda} \approx 1ns$$

relaxation time, for ions movement in solution



(single) ion channels, capacitive properties individual ions…

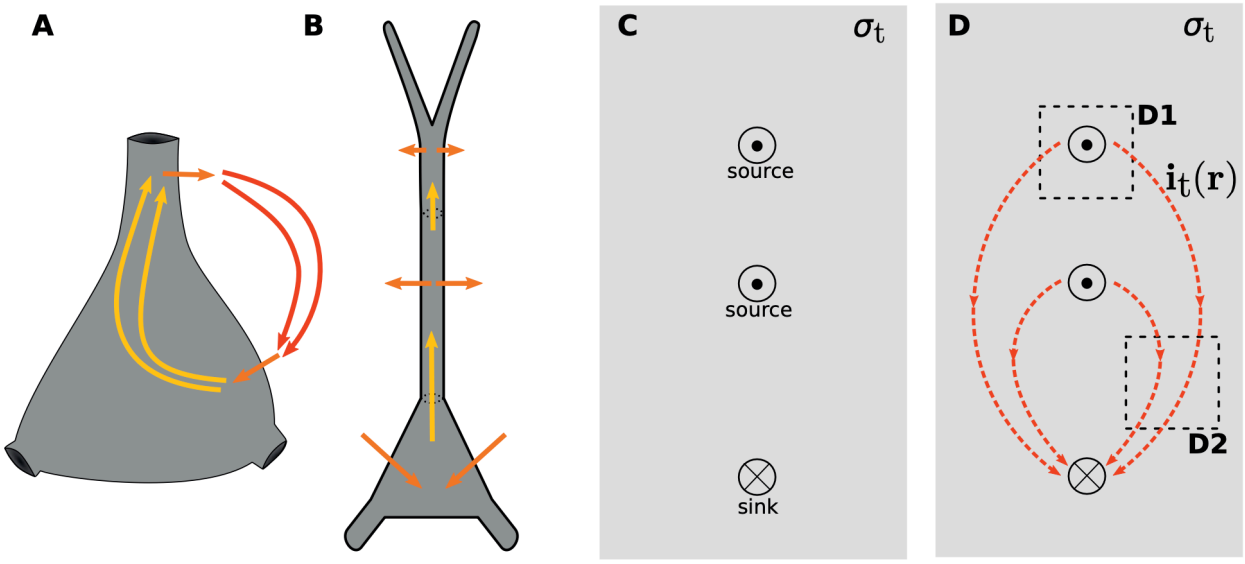
concentrations, densities,…

bulk properties, current flows

At ***macroscale*** (i.e. distances >> Debye's and times >> relaxation time) intracell. and extracell. spaces are effectively **electroneutral (thus isopotential)**

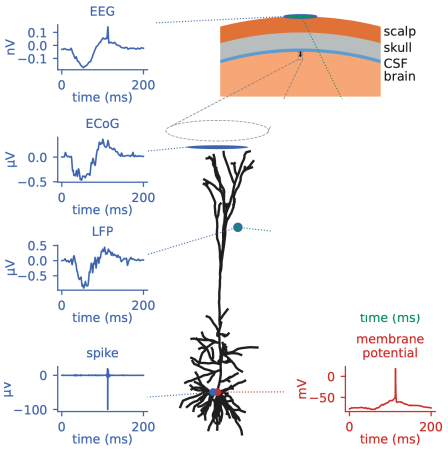
Extracellular potentials can be inferred (in much simpler terms than ionic charges distribution and movements) from the **conservation of charge**, **applied to currents**: *no accumulation of current exist (Kirchoff's).*

Not "charges" but "*Current sources (and sinks)*"



Halnes *et al.*, 2024

*Extra*cellular potential *Ve*
Volume Conductor Theory



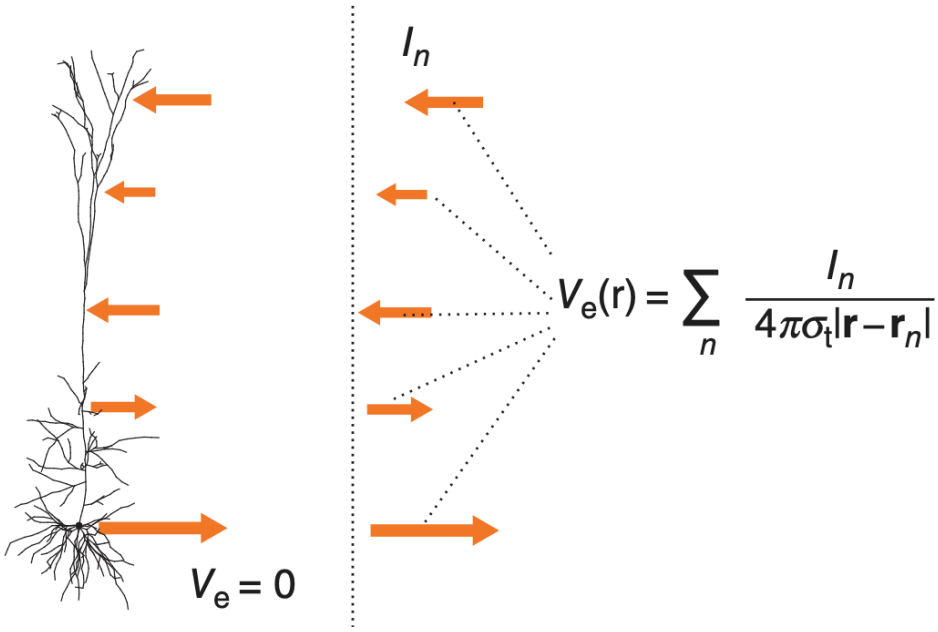
Halnes *et al.*, 2024

Assumptions of VCT:

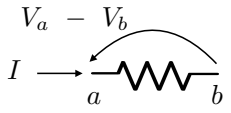
- all intracellular *V(x,t)* and *I(x,t)* are known or measured;
  - *(from cable theory, cell excitability, synapses location)*
- *Ve* does NOT affect intracellular *V* and *I (no "ephaptic" interactions)*
- the brain tissue is a continuous medium (i.e. a "volume" conductor)
- the macroscopic tissue conductivity is uniform (in space), constant (in time), and isotropic (across space directions);
- at macro scale the tissue is Ohmic (in 3D)

Starting point for "forward" description of
the origin of extracellular spikes, LFPs, ECoGs, EEGs.

*Extra*cellular potential *Ve*
Volume Conductor Theory



Ohm's law (for a volume c.) and *point* sources

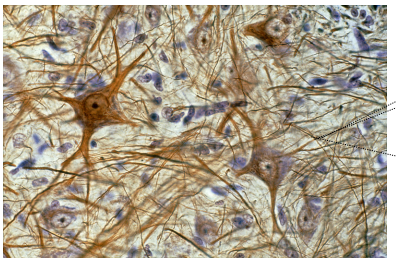


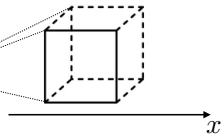
$$I = -G\,\Delta V$$

current

$$I = -\sigma_t \frac{S}{\Delta x}\Delta V \qquad i_t\,S = -\sigma_t \frac{S}{\Delta x}\Delta V$$

$$i_t = -\sigma_t \frac{d}{dx} V$$



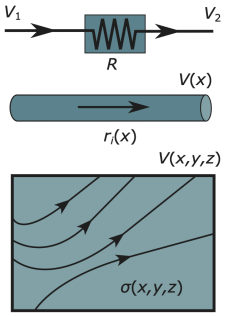


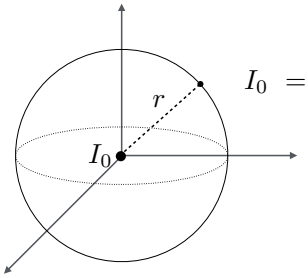
$$\Delta V = V(x+\Delta x) - V(x)$$

$$i_t = -\sigma_t \nabla V_e$$

current **density** through the **tissue** | macroscopic **tissue** conductivity | gradient of extracellular electrical potential

$$i_t = \sigma_t \mathbf{E}$$





$$I_0 = 4\,\pi r^2\, i_t(r)$$

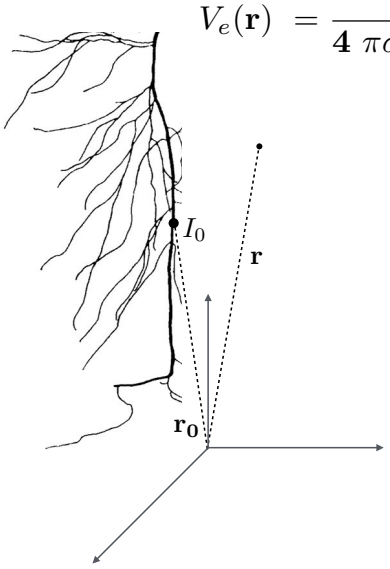
$$\frac{I_0}{4\,\pi r^2} = -\sigma_t \frac{d}{dr} V$$

$$\int_\infty^r \frac{I_0}{4\,\pi r^2}\,dr = -\int_\infty^r \sigma_t \frac{d}{dr} V\,dr$$

$$-\frac{I_0}{4\,\pi r} + 0 = -\sigma_t\, V(r) + 0$$

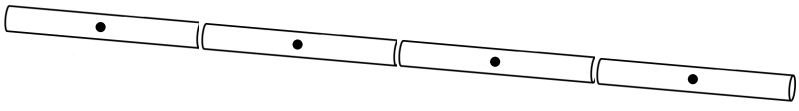
(A set of) "Point current" source(s)

$$V_e(\mathbf{r}) = \frac{\mathbf{I_0}}{\mathbf{4}\,\pi\sigma_\mathbf{t}||\mathbf{r}-\mathbf{r_0}||}$$

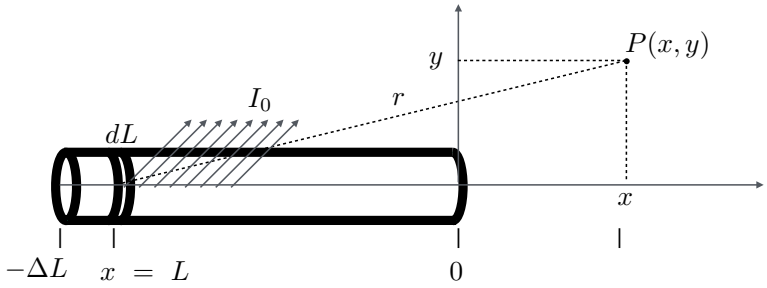


$$V_e(\mathbf{r}) = \frac{I_1}{4\pi\sigma_t\|\mathbf{r}-\mathbf{r}_1\|} + \frac{I_2}{4\pi\sigma_t\|\mathbf{r}-\mathbf{r}_2\|} + \frac{I_3}{4\pi\sigma_t\|\mathbf{r}-\mathbf{r}_3\|} + \ldots = \sum_n \frac{I_n}{4\pi\sigma_t\|\mathbf{r}-\mathbf{r}_n\|}$$

Point-source approximation: a neuron = a collection of point-current sources



A **better** approximation: “line”-sources



$$r = \sqrt{(x-L)^2 + y^2}$$

$$V_e(\mathbf{r}) = \frac{\mathbf{I_0}}{\mathbf{4}\,\pi\sigma_\mathbf{t}||\mathbf{r}-\mathbf{r_0}||}$$

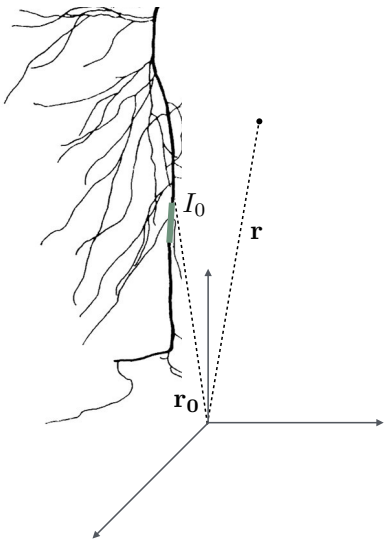
$$dI_0 = I_0\,\frac{dL}{\Delta L} \qquad V_e(\mathbf{r})_{dL} = \frac{I_0}{\Delta L}\,\frac{dL}{4\,\pi\,\sigma_t\,\sqrt{(x-L)^2+y^2}} \qquad V_e(\mathbf{r}) = \int_{-\Delta L}^{0} \frac{I_0}{\Delta L}\,\frac{dL}{4\,\pi\,\sigma_t\,\sqrt{(x-L)^2+y^2}}$$

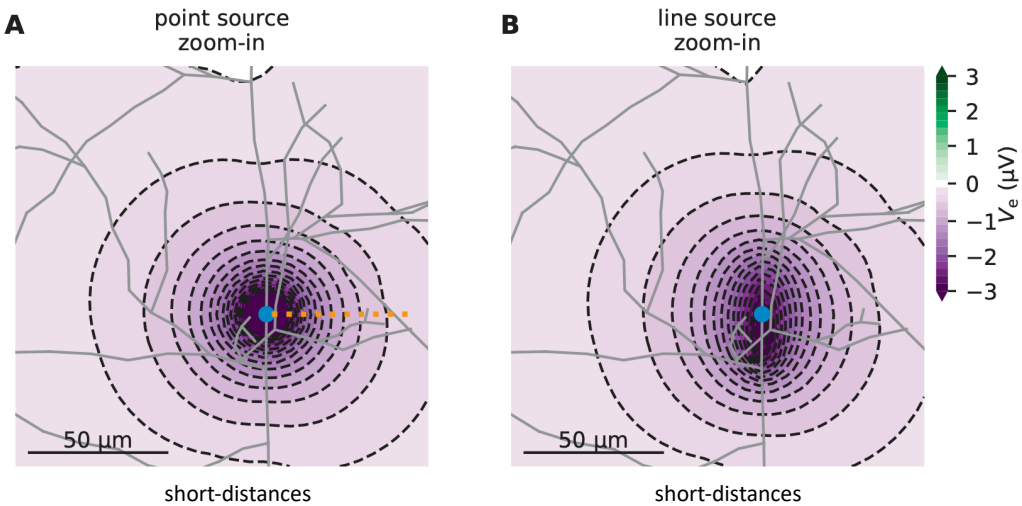
$$\int \frac{dt}{\sqrt{t^2+a^2}} = \ln\left|t+\sqrt{t^2+a^2}\right| + K$$

$$V_e(\mathbf{r}) = \frac{I_0}{4\,\pi\,\sigma_t\,\Delta L}\,\ln\frac{\left|-x+\sqrt{x^2+y^2}\right|}{\left|-(x+\Delta L)+\sqrt{(x+\Delta L)^2+y^2}\right|}$$

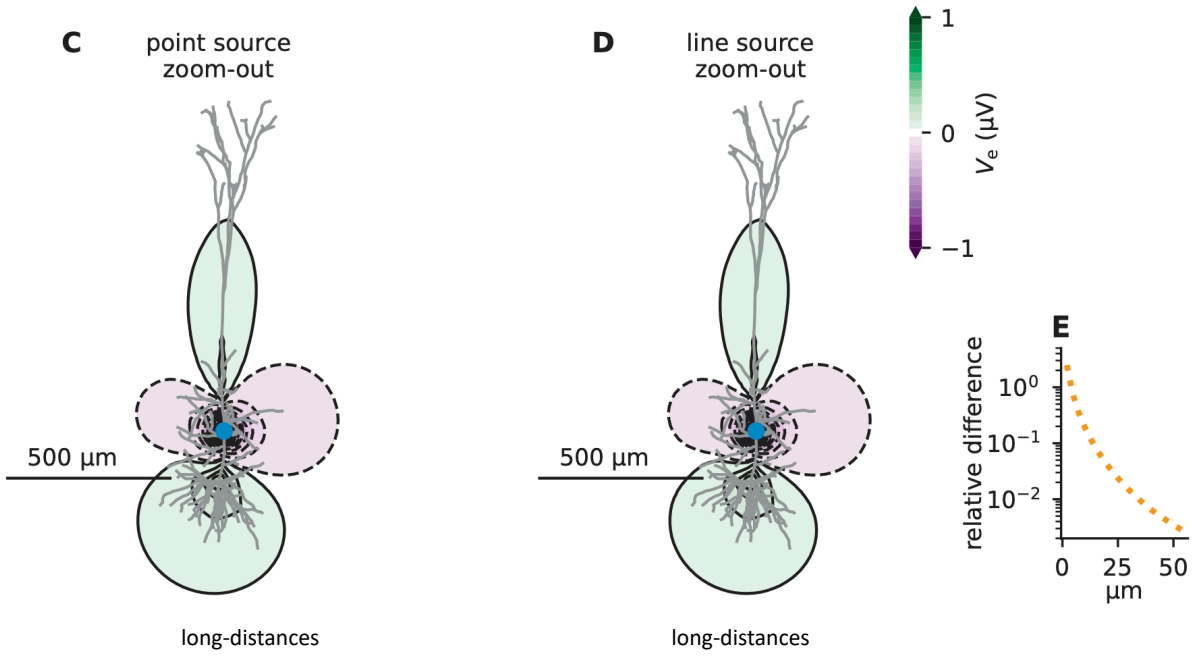
“Point-source” vs “Line-source” approaches

$$V_e(\mathbf{r}) = \frac{I_0}{4\,\pi\,\sigma_t\,\Delta L}\,\ln\frac{\left|-x+\sqrt{x^2+y^2}\right|}{\left|-(x+\Delta L)+\sqrt{(x+\Delta L)^2+y^2}\right|}$$





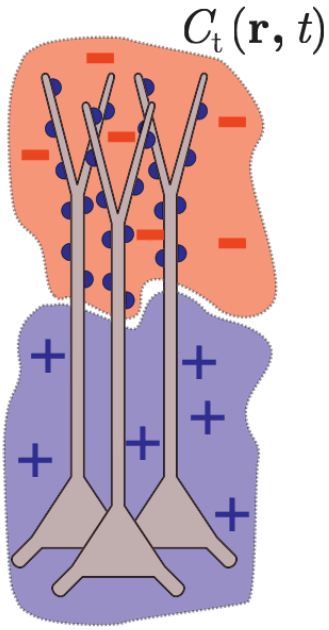
“Point-source” vs “Line-source” approaches



ATTENDANCE TRACKING - **code RBZ2M**
(for statistical purposes only)

https://www.unimore.it/it/servizi/unimore-app

More generally: CSD - current source *density*



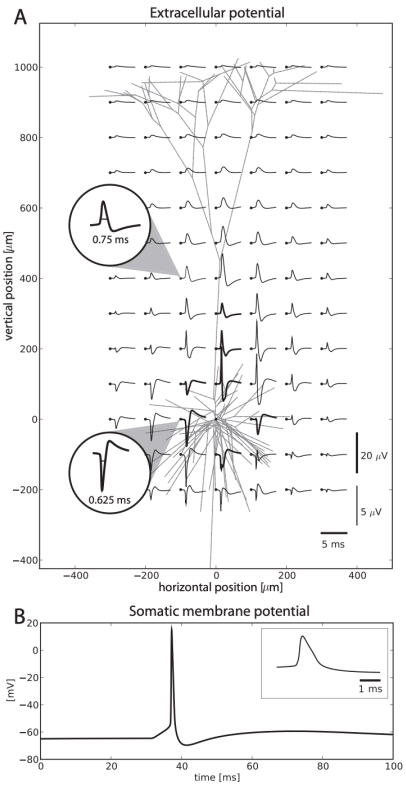
- current source density (in the tissue) Ct is
  - *a continuous function of time and position*
  - *usually estimated in the "inverse" problem (from measured Ve to Ct)*
  - *can be a collection of Dirac's delta functions (in 3D), and then it reduces to the point-source approximation already discussed*

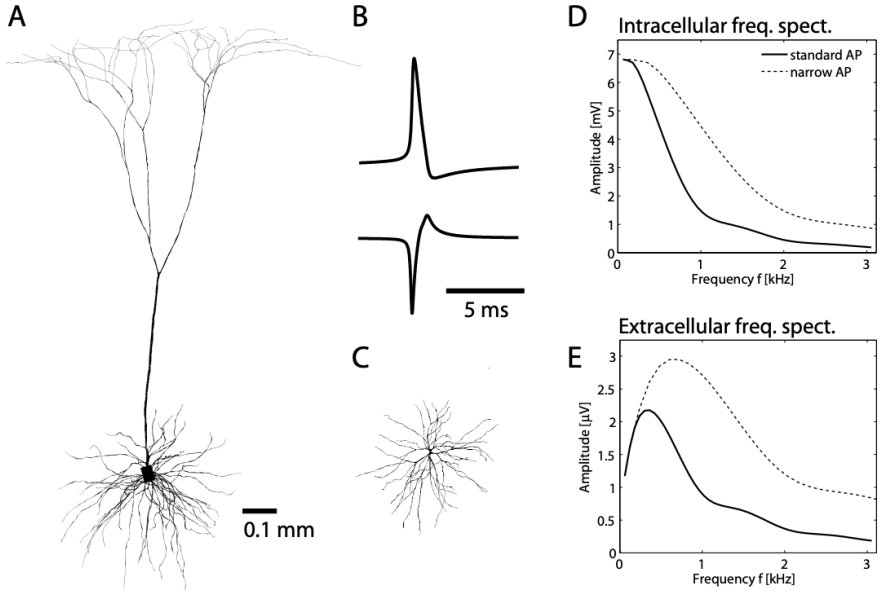
$$\nabla\ i_t\ = C_t$$

$$\sigma_t \nabla^2 V_e = -C_t$$

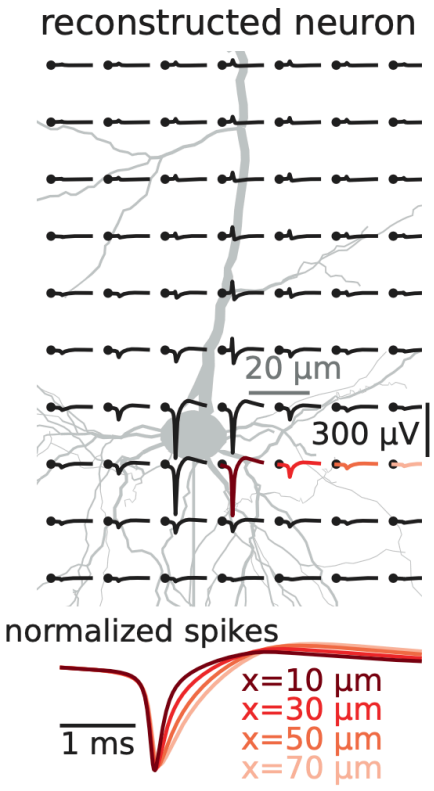
More complicated partial-derivatives differential equation…

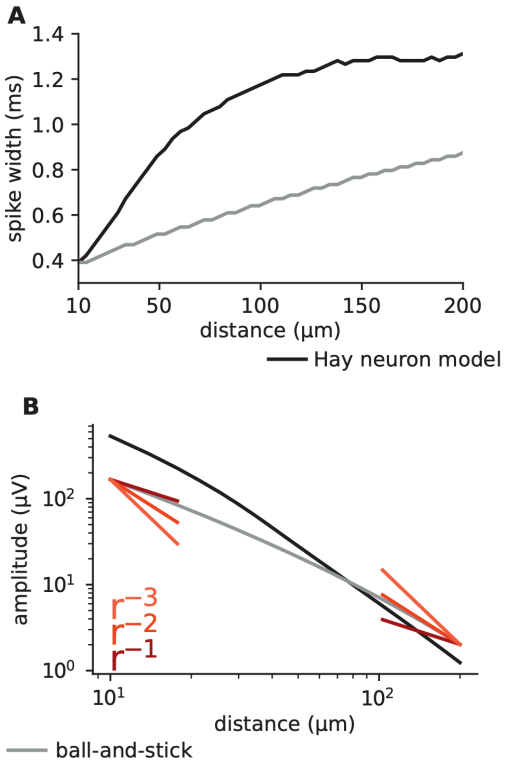
VCT - spikes



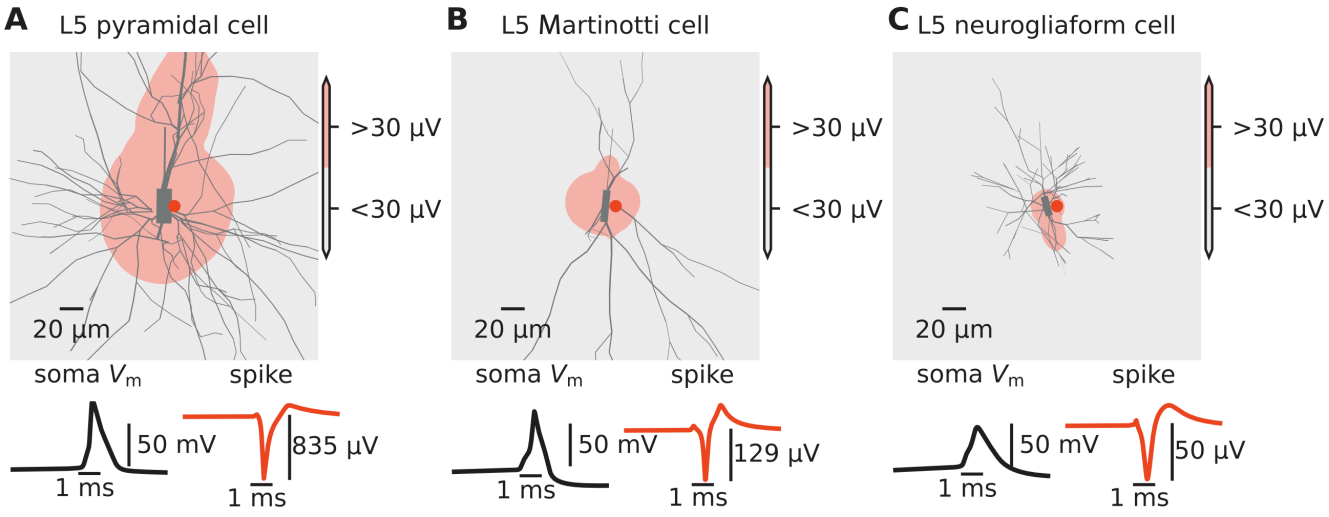


Spike shapes

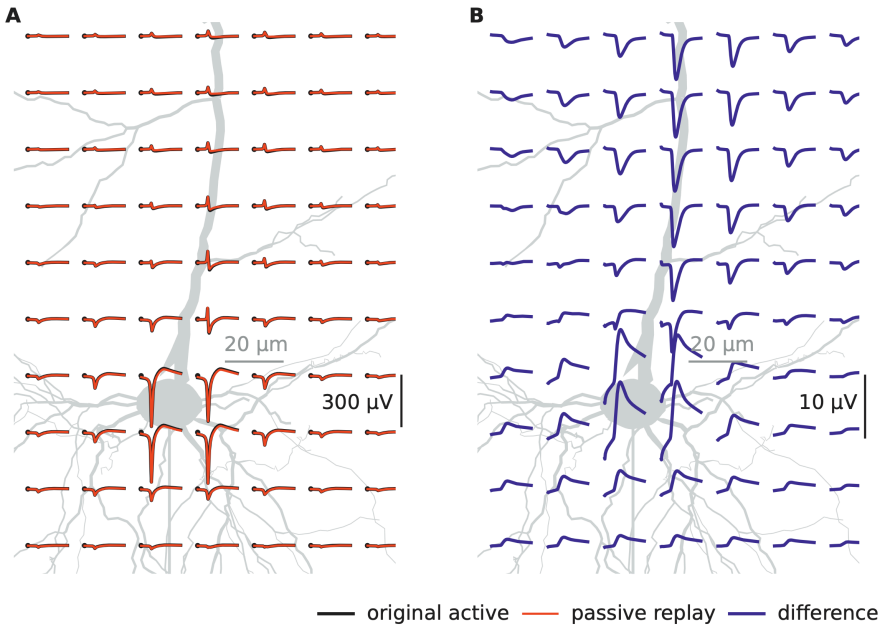




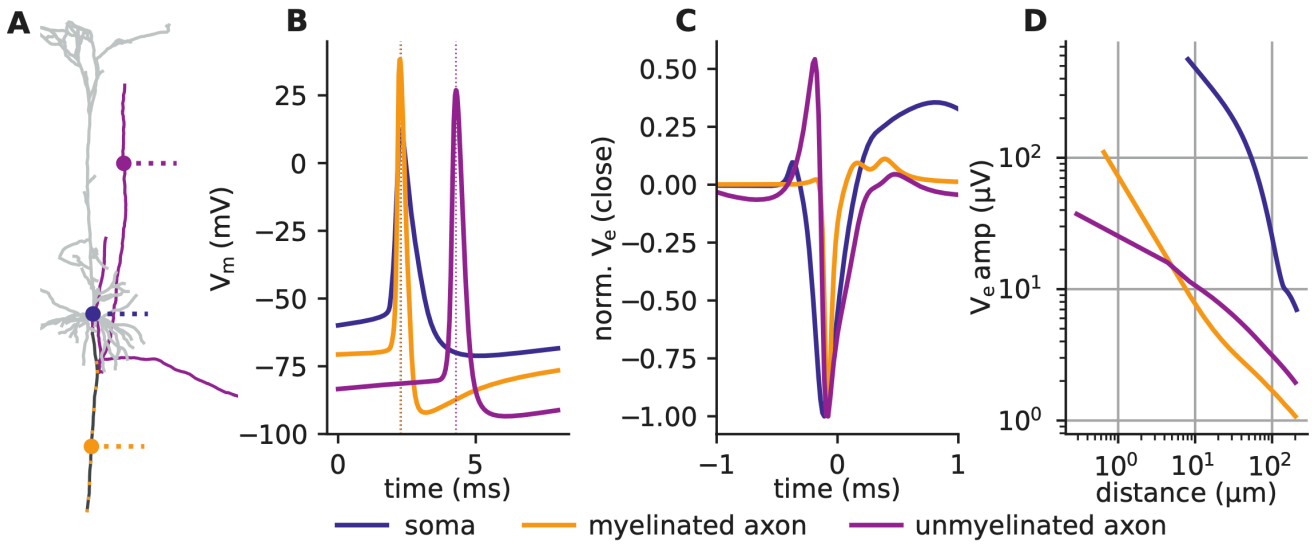
Spike detectability: depends on cell morphology

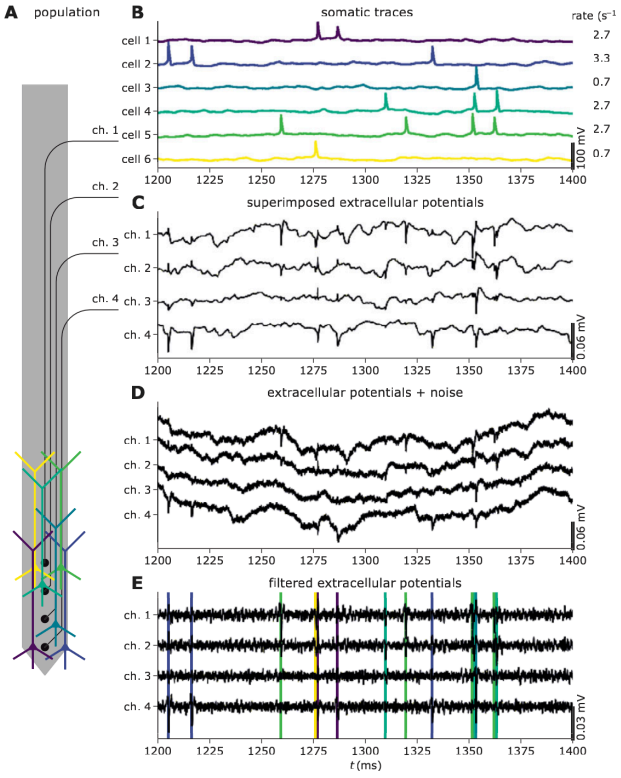


Spikes: weak impact of *active* dendrites



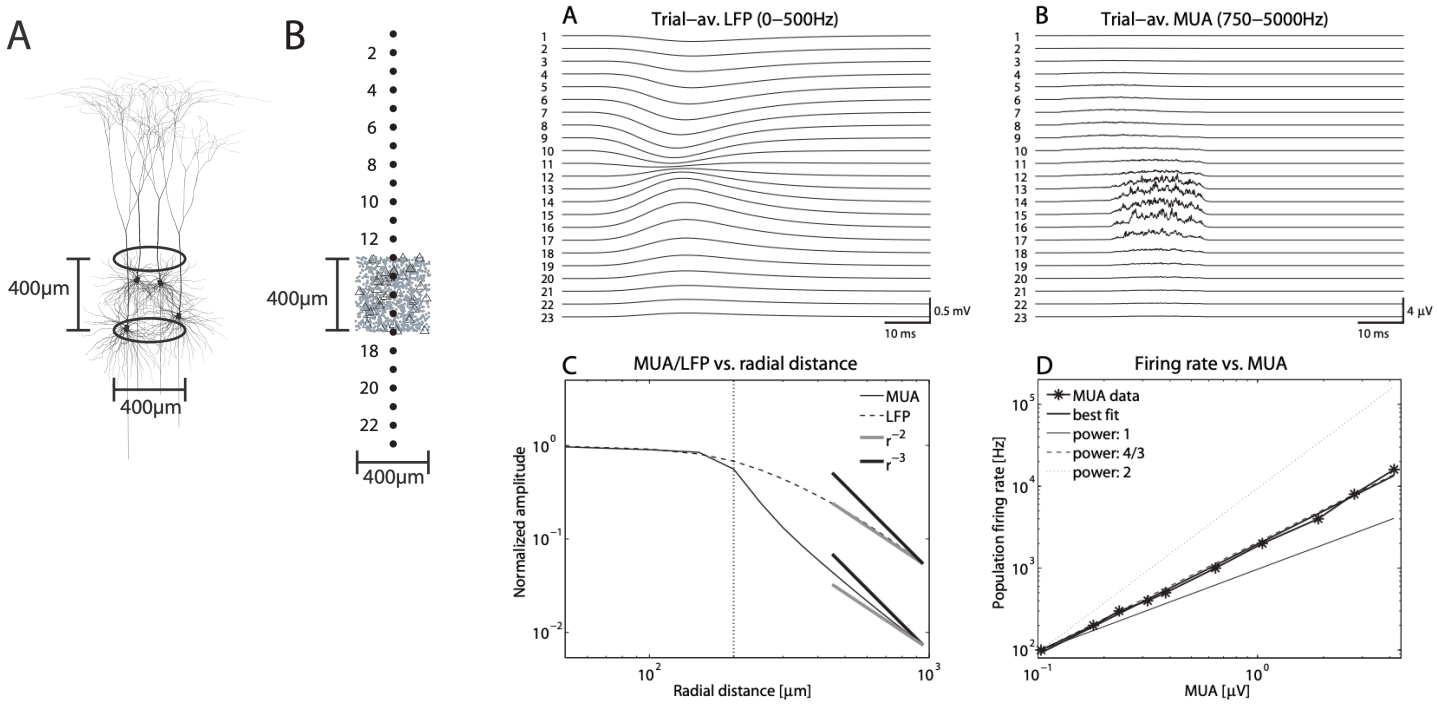
Axonal Spikes: impact of *myelinated* axon





Forward modeling of Ve may guide development/benchmark of *spike-sorting* algorithms.

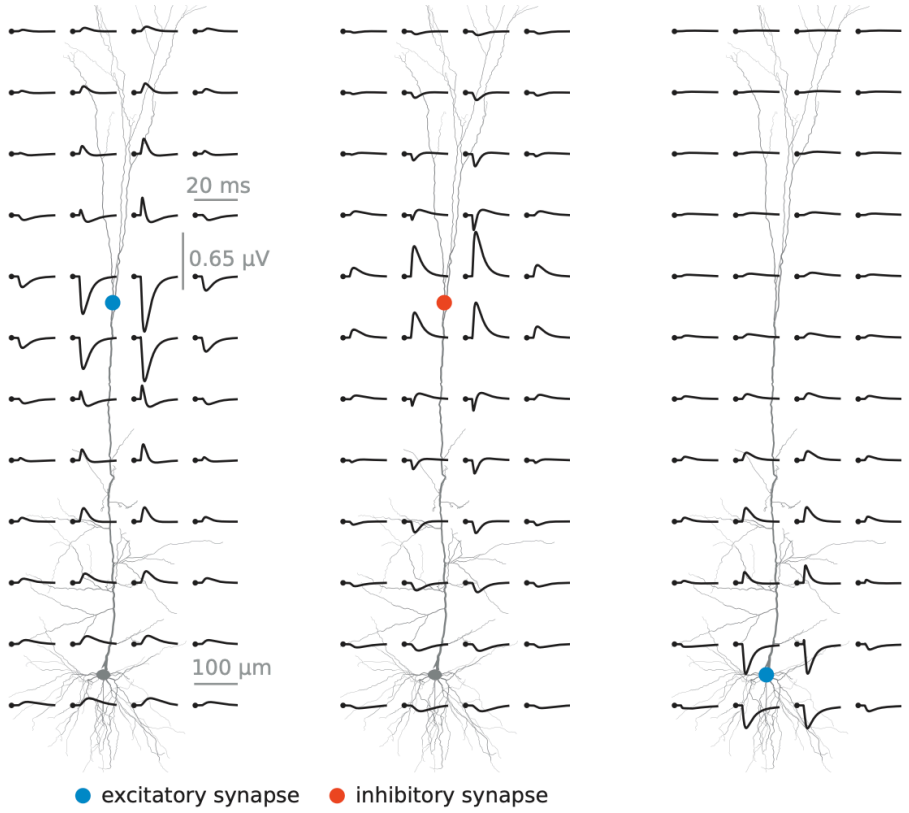
VCT - LFP and MUA - population of pyr. cells



Local Field Potentials

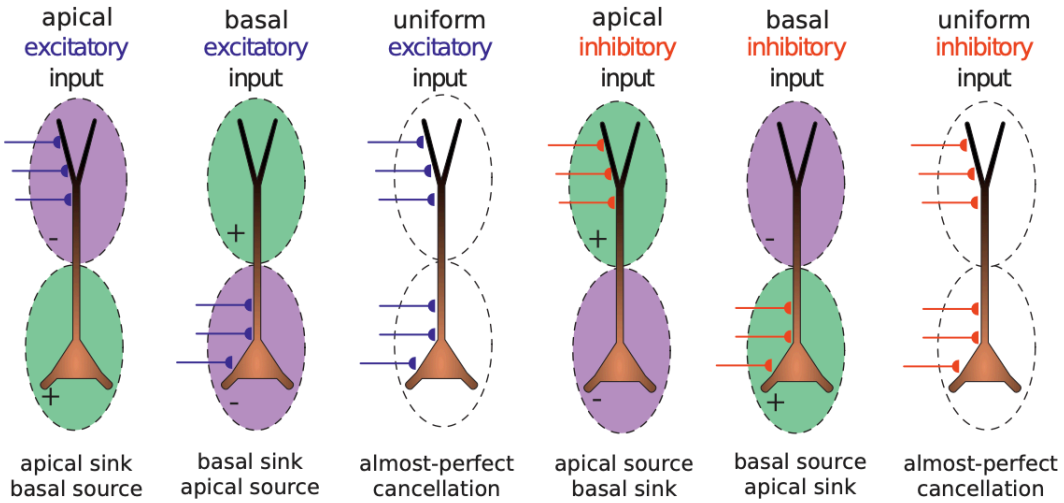
Several processes contribute to LFP (the low-frequency part of the extracell.V)

- **synaptic currents (and associated return currents) - MAIN CONTRIBUTOR**
- regular spikes, contributing to low-frequencies (e.g. sodium-mediated AP)
- slow intrinsic membrane currents, mediated by NMDAr or Ca-spikes, etc.
- membrane currents in glial cells
- current loops through neurons, glia and the extracellular space
- diffusion electrical potentials arising from concentration differences of ions in the extracellular space.



Typical “motifs” of configuration:

sources an sinks’ arrangement is not “obvious”



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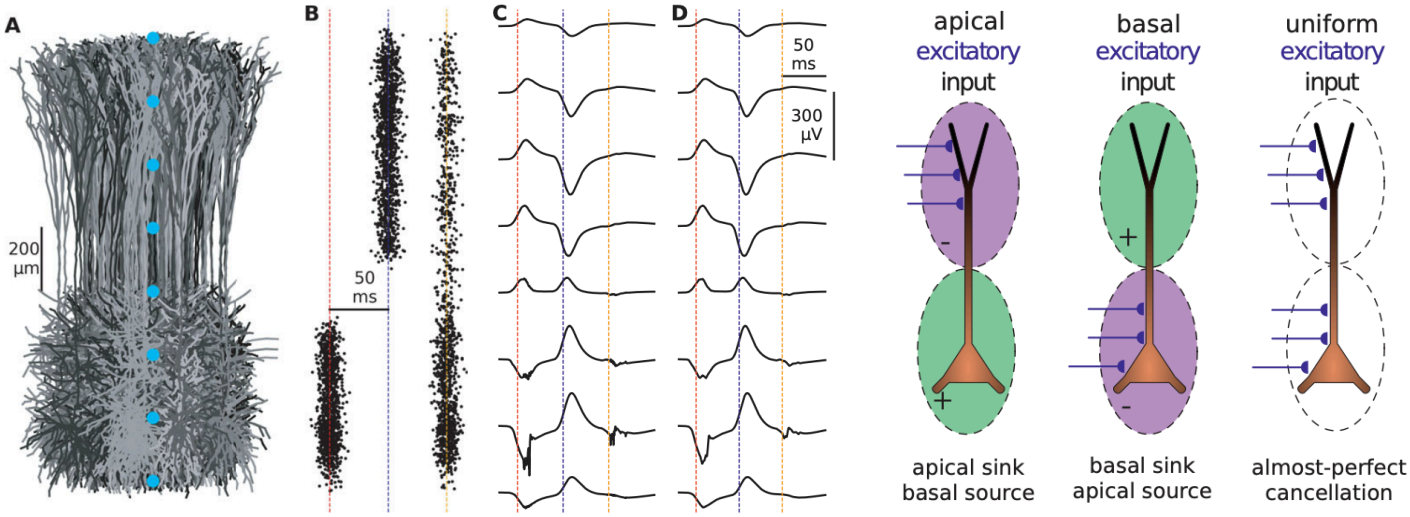
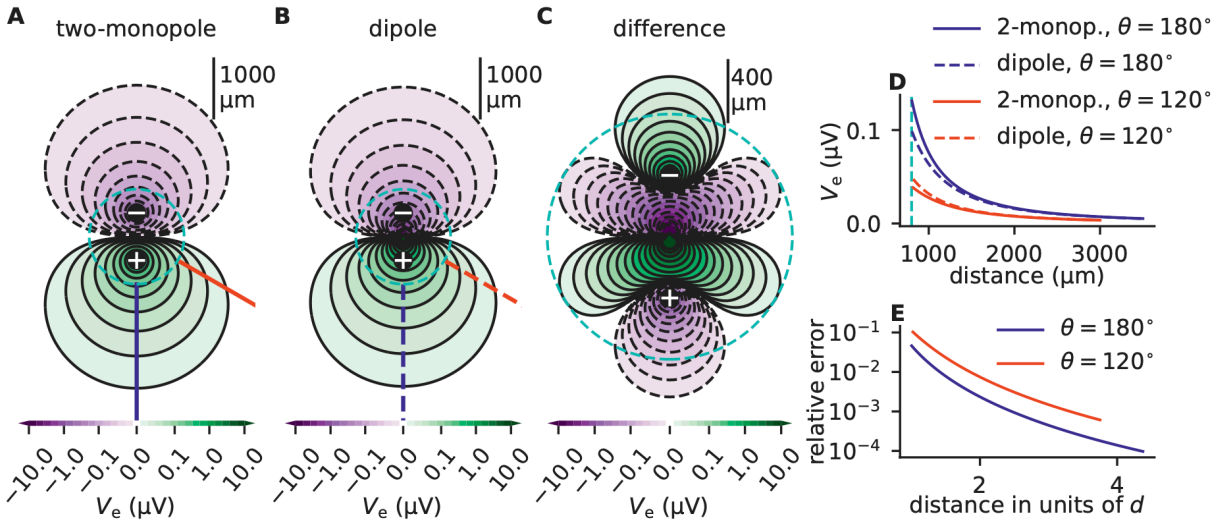


At large distance extracellular potentials resembles those generated by a *dipole*



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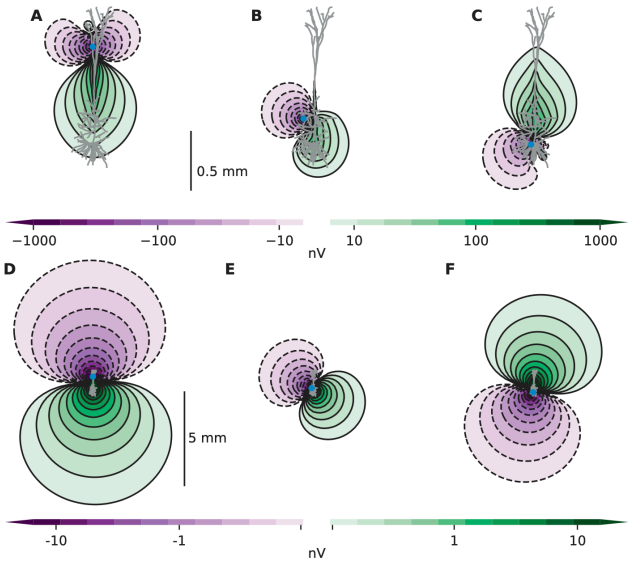

$$V_e(r) \approx \frac{1}{4\pi\sigma_t}\frac{P\ cos\theta}{R^2}$$

$$P = \sum_{n=1}^{N} I_n\ \mathbf{r_n}$$

$$P = -I\ \mathbf{r_1} + I\ \mathbf{r_2} = I\ \mathbf{d}$$